



SUPERCLOUD COMPOSER® ARCHITECTURAL FRAMEWORK

Defines how modern data centers are designed, deployed and operated incorporating AI, HPC, Liquid-Cooled and Enterprise.

TABLE OF CONTENTS

Executive Summary.....	1
Why SuperCloud Composer?.....	3
SuperCloud Composer Core Functionality and Features.....	8
Conclusion.....	38

Executive Summary

Today's datacenter requirements are under pressure to reduce costs, whilst ensuring operational efficiency and agility. Infrastructure requirements must be able to scale across various technologies, enterprise, AI, and high-performance workloads without introducing complexities and additional overheads. IT leaders are increasingly adopting time-bound, workload-driven infrastructure

consumption to complement existing fixed-purpose systems. In parallel, responsibility for cost control has expanded beyond compute and software into energy and cooling. Power consumption, thermal density, and cooling efficiency. Once largely fixed operational inputs are now highly variable and workload-dependent, particularly with GPU-dense and liquid-cooled environments. This shift places additional operational and financial burden on CTOs and IT leadership, who must now actively manage and optimize infrastructure not just for performance and availability, but for energy efficiency, cooling utilization, and total cost of ownership in real time.

SuperCloud Composer provides a unified management and orchestration platform spanning Supermicro infrastructure for multiple generations of infrastructure technology.

Digital transformation, continuous innovation, and sustained data growth are increasing the cost and frequency of infrastructure refresh and scale-out. Despite this investment, data-center resources remain underutilized, operational efficiency remains limited, PUE continues to rise, physical footprints expand, and operational staffing costs increase. Together, these pressures are challenging the economic sustainability of traditional data-center models.

As CPU and GPU TDP (Thermal Design Power) continue to increase, liquid cooling is becoming a foundational technology for modern data centers. At higher power densities, air cooling alone is increasingly inefficient and costly. Liquid-cooled environments introduce additional infrastructure components, spanning servers, CDUs, and facility-level cooling systems that require consistent monitoring, telemetry, and operational control.

Traditional infrastructure models rely on rigid hardware configurations, slow provisioning processes, and fragmented management tools. This approach limits visibility, analytics, and lifecycle automation, and lacks a unified platform capable of managing AI, HPC, Enterprise and liquid-cooled environments at scale.

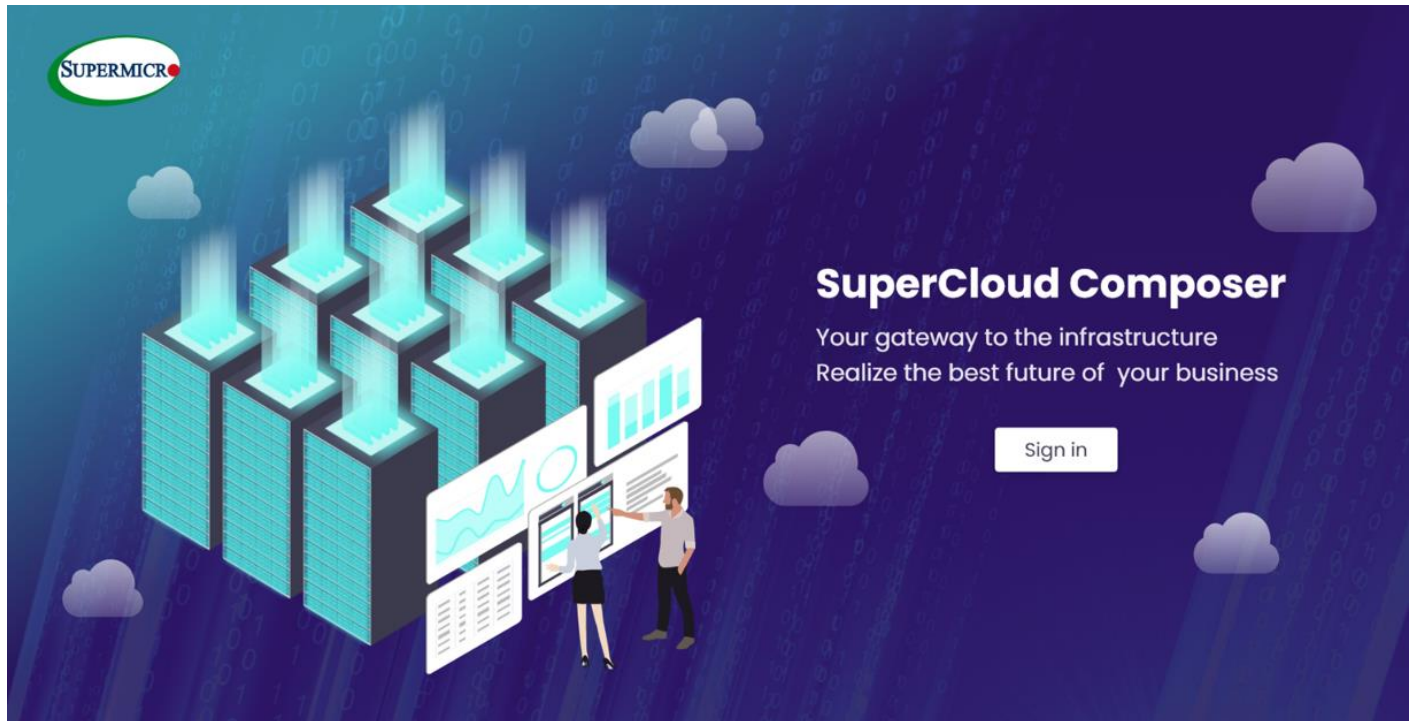


Figure 1. SuperCloud Composer Landing Page

SuperCloud Composer addresses these challenges by providing a unified architectural framework for designing, deploying, and operating modern data centers across heterogeneous infrastructure and workload domains

Delivering centralized visibility, telemetry, and lifecycle automation across infrastructure, enabling higher utilization, improved operational efficiency, and consistent control as environments scale and evolve.

Supermicro's approach is centered on ensuring that the data center operates as an integrated, efficient system rather than a collection of isolated components. SuperCloud Composer (SCC) is designed to provide end-to-end visibility and control across infrastructure health, extending beyond server management to include power delivery and cooling systems. This includes emphasis on power usage awareness, active CDU monitoring, and real-time infrastructure telemetry, enabling operators to understand, optimize, and safely scale high-density environments.

By aligning system health, power consumption, and cooling behavior within a single control plane, SCC supports more predictable operations, improved efficiency, and informed infrastructure decision-making.

Why SuperCloud Composer?

- True Single Pane of Glass with a consistent, intuitive management experience
- Unified visibility across Compute, Storage, Networking, and Rack Infrastructure
- Standards-based Redfish northbound API for integration and automation
- Redfish BMC monitoring and alerting for system and hardware components
- Day-0 to Day-2 lifecycle management: discovery, provisioning, firmware and compliance
- Multi-generation Supermicro support enabling mixed compute fleets
- Liquid-cooling monitoring & awareness, including In-Rack, In-Row, Sidecar and RDHx
- CDU monitoring with policy-based enforcement to detect cooling faults and trigger protective actions
- Built to scale from single racks to Enterprise, AI and HPC clusters
- Role-based access control aligned to platform and SRE operating models
- Detailed, Real-Time Telemetry for hardware health and performance trends



Figure 2. Monitor, Manage, Update & Maintain (infographic)

SuperCloud Composer Deployment Domains

- **Cloud, Neo-Cloud, Hyperscale, HPC and AI Data Centers**
 - Supports large-scale environments by providing centralized visibility and operational control for high-density AI and GPU infrastructure, with a strong focus on firmware lifecycle management, telemetry, and cooling reliability.
- **Enterprise Data Centers**
 - In enterprise environments, SuperCloud Composer is deployed alongside traditionally operated infrastructure to deliver unified monitoring, firmware governance, and operational lifecycle management across compute, accelerators, and liquid-cooled systems.
- **Edge & IoT Deployments**
 - Offers centralized visibility and control for distributed edge systems, maintaining firmware consistency, health status, and cooling awareness across geographically dispersed locations.



Figure 3. Market Deployment (infographic)

Target Verticals

A single, consistent infrastructure and monitoring model supports a wide range of industries, enabling unified visibility, telemetry, and operational control across diverse workloads.

- 30+ years of systems expertise across enterprise, HPC, AI and mission-critical workloads.



Figure 4. Commercial Sector Experience (infographic)

Single Pane of Glass

- Single source of truth for all infrastructure.
- Operational control – Compliance and Governance.
- State & health awareness.
- Real-time telemetry.
- Policy driven alerting.

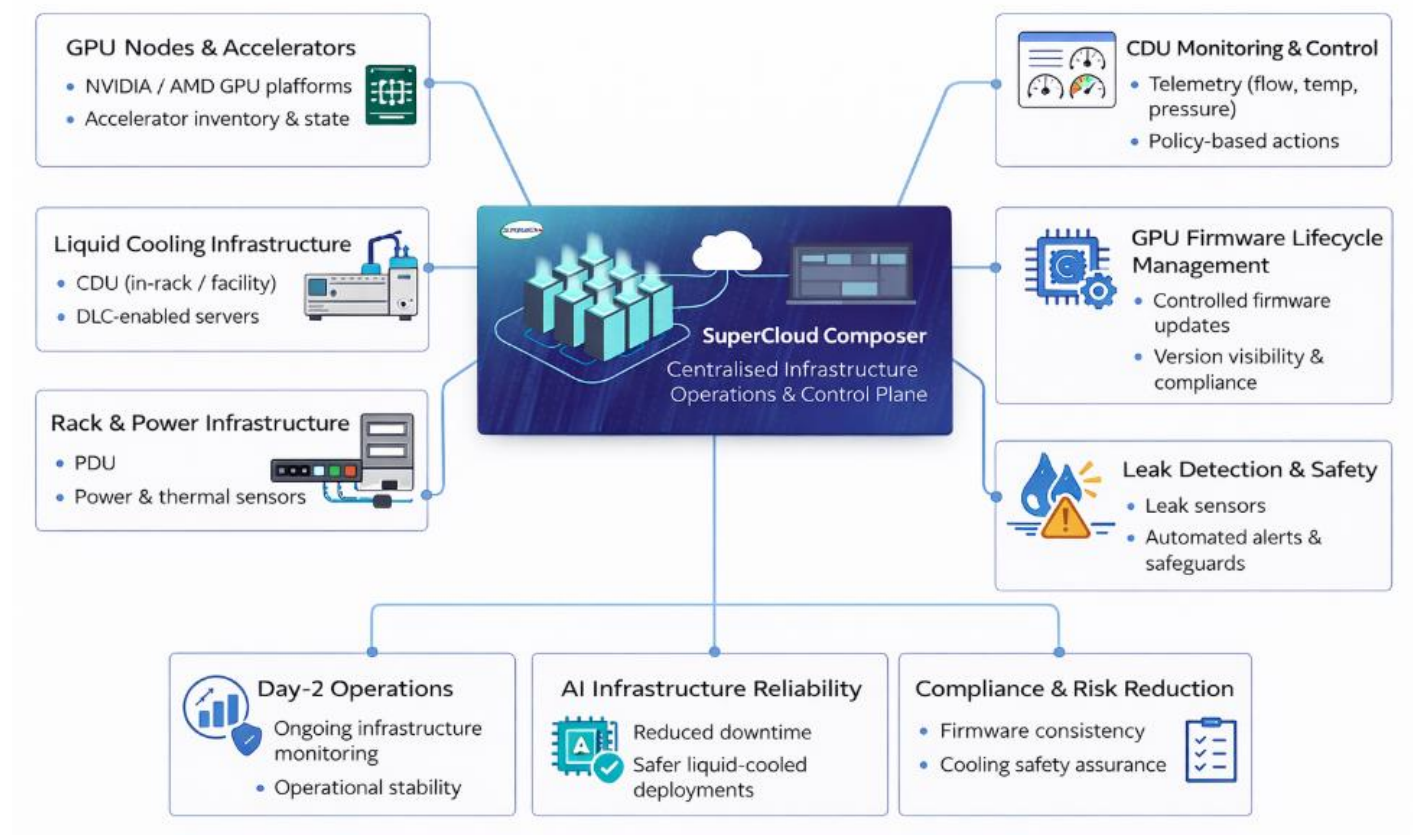


Figure 5. Single Pane of Glass Observability & Management (infographic)

Unified Monitoring

- Scales seamlessly from a single high-value rack to multi-site data centers through one unified control plane.
- Targeted access to any device—BMC, IPMI, network, power and cooling across environments of any size.
- Single datacenter tool of Operations.

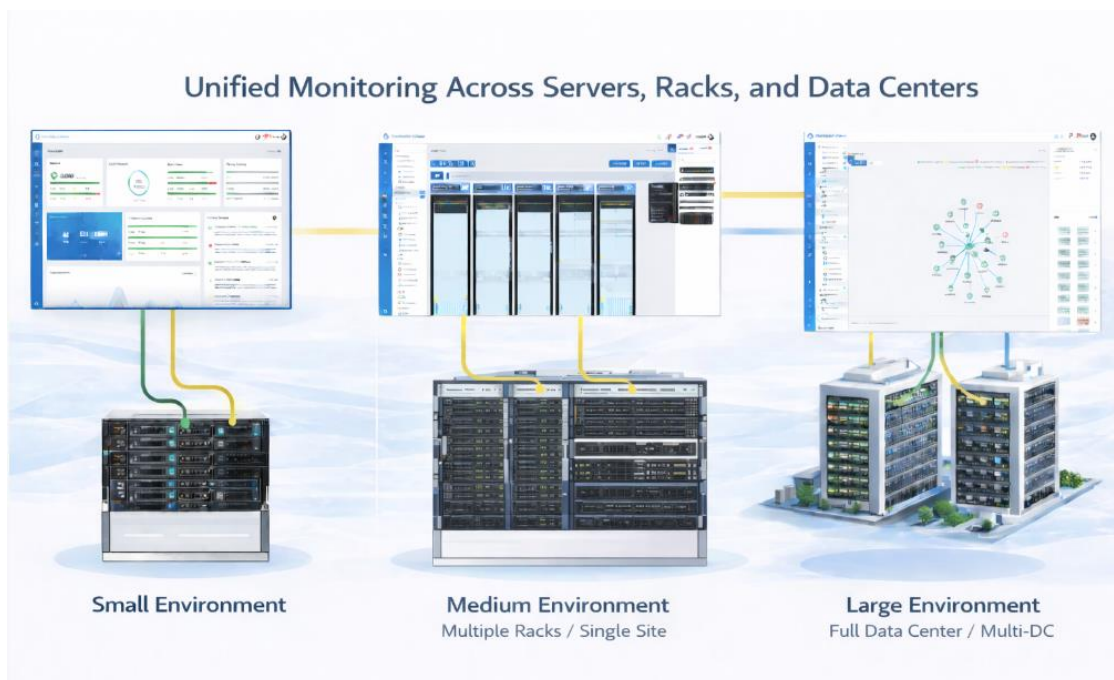


Figure 6. Unified Monitoring Across Multi-Site and Multi-Region Environments, from Edge to Hyperscale (infographic)

SuperCloud Composer Core Functionality and Features

SuperCloud Composer is a centralized infrastructure management platform designed to support both traditional enterprise server environments and modern AI infrastructure. It provides a unified control plane for infrastructure operations and lifecycle management, delivering operational visibility and control across compute platforms as well as GPU-accelerated and liquid-cooled systems. Built on open standards such as Redfish, SuperCloud Composer is delivered through a modular, web-based interface designed for scalable, day-to-day operations.

- SuperCloud Composer provides guided installation experience, supporting CLI and GUI driven setup to simplify deployments.
- SuperCloud Composer can be deployed as a virtual appliance or on bare metal, on Linux operating systems and supporting high-availability installation.
- Provides a single control plane for instantiating, monitoring and managing infrastructure components with a focus on GPU Platforms, power and cooling infrastructure.
- SuperCloud Composer leverages Redfish and open management interfaces to integrate with supported hardware and non Supermicro infrastructure, enabling extensibility without vendor lock-in.
- SuperCloud Composer is tightly integrated with CDUs, DLC-enabled servers, and leak detection systems, delivering telemetry, safety monitoring, and policy-based actions to improve reliability in liquid-cooled AI deployments.
- SuperCloud Composer provides centralized management and monitoring of PDUs, delivering real-time power telemetry, outlet-level control, and policy-driven actions integrated with compute and cooling operations.
- SuperCloud Composer acts as a single source of truth for BMC, IPMI, and field-replaceable (FRU) component data, enabling secure, direct access and lifecycle visibility across thousands of compute, storage, network, power, and cooling devices from one interface.
- SuperCloud Composer enables remote, at-scale firmware updates for system and GPU components, supporting consistent lifecycle management across dispersed sites and large infrastructure footprints.

Discovery & Inventory	Provisioning & Configuration	Monitoring & Telemetry	Operations & Control	Power Management	Cooling Management
Physical Asset Discovery	Baremetal OS Provisioning	Compute, GPU, Storage Metrics	Single-Pane Device Access (BMC, PDU, CDU)	Grid Management	CDU Monitoring & Control
BMC / Redfish / IPMI Access	Network Provisioning (Switch & Fabric)	Power, Thermal & Cooling Telemetry	Topology & Dependency Visualization	Energy & Capacity Visibility	Leak Detection & Safety
Physical Location & Rack Position	Firmware Lifecycle Management	Time-Series Analytics & Dashboards	POD View & Rack Management	PDU Outlet & Power Path Mapping	Policy Driven Leak Actions
GPU & Accelerator Inventory	Policy-Based Configuration	Alerting & Threshold Policies	Northbound APIs Swagger-Based Documentation & Redfish OpenAPI Support	Circuit & Breaker Visibility	CDU Firmware Update Capability
Network Switch Port Inventory	Parallel (Multi-Device) Deployment	System & Component Health Status	Role-Based Access Control (RBAC)	Power Redundancy Validation (A/B feeds)	Support for In-Rack, In-Row, RDHx, Sidecar & Cooling Tower
Firmware & BIOS Inventory	Scale to 20k Hosts	Capacity Threshold Forecasting	ServiceNow Integration	Policy-Driven Power Actions	Cooling Capacity Forecasting
Rack, PDU, CDU, Switch Discovery	Air Gapped Installation	Event Logging & Alerting	Single-Pane Device Access (BMC, PDU, CDU)	Thermal Hotspot Identification	Coolant Flow, Pressure, Temperature Telemetry
Asset Metadata & Tagging	BareMetal & VM Deployment	Performance Metric - Telemetry Out of Band & In Band	Multi-Site Operations	Power Usage Effectiveness (PUE) Inputs	Cooling Loop Dependency Mapping
Commissioning & Decommissioning	Dedicated Installation & Support By Expert Engineers	Multi-Channel Alerting via Webhooks, Email, Teams & Slack	Call Home Service - Support Advisor	Utility Feed Ingress Monitoring (MW / kVA)	Cooling Maintenance State & Bypass Awareness

Figure 7. Features and Capabilities

Unified Web Based Management Interface

- Web-based management interface, optimized for modern browsers (Chrome, Edge and Firefox), and designed around consistent operational best practices to simplify day-to-day infrastructure management.
- Built-in analytics and visualization deliver system-level insight into compute, IOPS, network performance, telemetry, thermal and power behavior, composed node health, storage utilization, and overall platform status.
- OS provisioning Allowing operating system deployment with support for metadata injection during build, meeting the needs of fast-moving DevOps environments.
- The Web UI is backed by Redfish-based RESTful APIs, exposing platform functionality programmatically through standardized, open management secure interfaces.
- The management platform supports high availability through a distributed Kubernetes-based architecture, ensuring continuous operation and automatic failover.

Standards-Based Redfish API for Secure Platform Integration

- SuperCloud Composer exposes platform functionality through Redfish-based RESTful APIs, including custom developed extended schemas that support SCC backend service agents and workflows.
- Standard Redfish schemas are supported, aligning with industry specifications to enable interoperability and integration with external tools and platforms.
- API interactions are secured using HTTPS, ensuring encrypted communication and enterprise-grade security.
- For more specific standardized Redfish API schemas, visit the site: <https://redfish.dmtf.org>

Identity, Access Control & Role Management

- Integration with enterprise identity providers including Keycloak, Active Directory and LDAP for centralized authentication, with a secure local administrative fallback to prevent platform lockout.
- Enforced password complexity, account lockout, and login rate limiting to align with enterprise security standards.
- Predefined user roles with scoped permissions, supporting full administration, infrastructure design, network configuration, and read-only access.
- Role-based separation of duties across compute, storage, network, and platform operations, applied consistently across the web interface and Redfish APIs.

Dashboard

The SuperCloud composer dashboard provides operational visibility into SCC platform health and a high-level overview of the managed environment's health and status through real time observations.

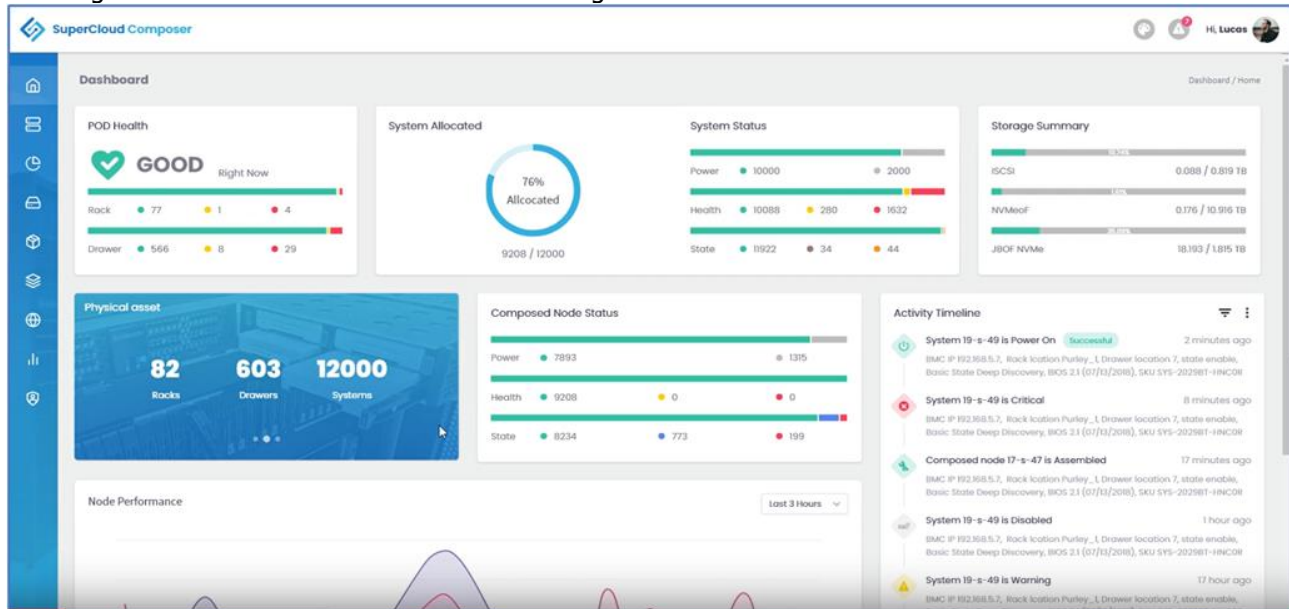


Figure 8. SCC Dashboard Landing Page

Interactive components enable administrators to drill down into detailed system, composed node, network and storage status directly from the dashboard.

Time-based charts provide flexible visibility into node performance, power consumption, and thermal behavior.

An integrated task and activity timeline provides visibility into ongoing and historical operations, including provisioning, configuration, and lifecycle actions.

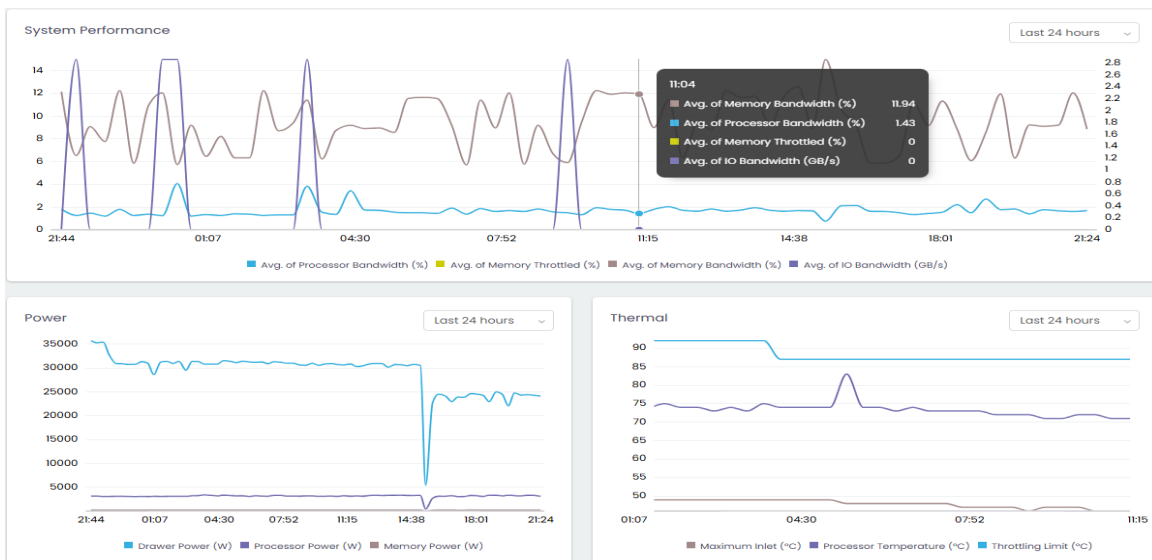


Figure 9. SCC Infrastructure Telemetry Dashboard

Pod View

POD View enables infrastructure to be organized and visualized using logical, POD-based groupings that reflect how systems are architected and operated. These groupings can align to physical layouts where required but are primarily designed to model operational domains such as workloads, tenants, power and cooling boundaries, or deployment units.

This provides operators with clear situational awareness to support deployment, operations, and ongoing infrastructure management.

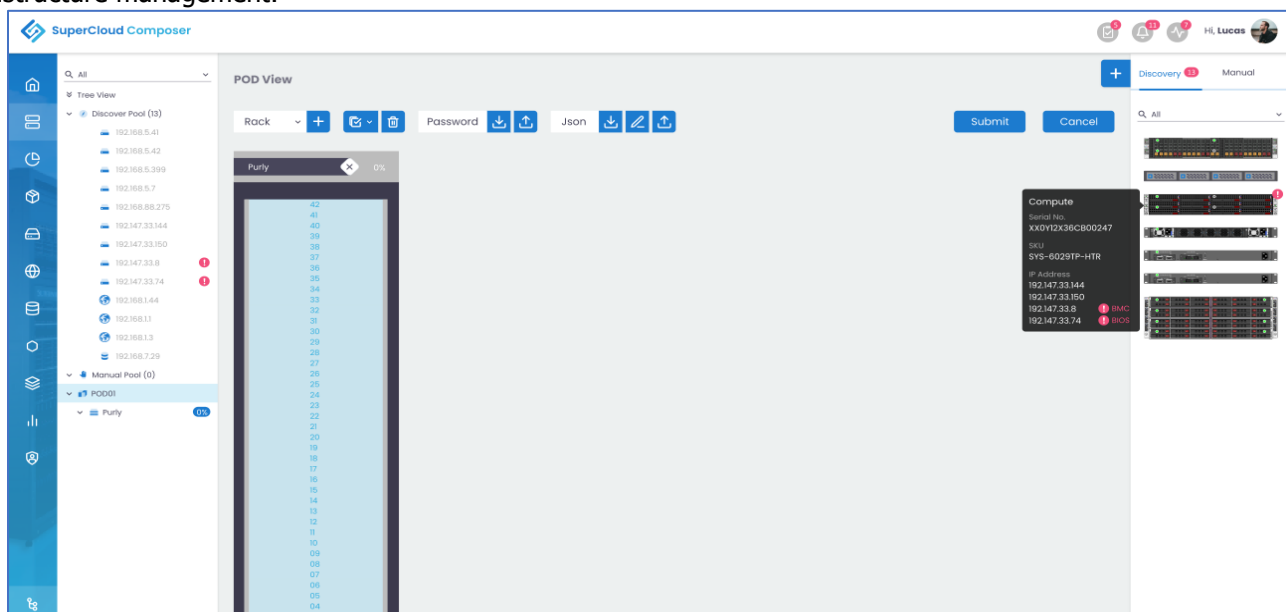


Figure 10. Pod View – Rack & Node Topology View

- Pod View provides a logical abstraction layer within SuperCloud Composer, enabling administrators to model the physical data center as workload-aligned pods composed of racks, drawers, and infrastructure resources.
- Physical systems are automatically discovered as they are powered on and presented in a discovery pool, allowing administrators to assign compute, storage, and network resources into pods through guided configuration workflows.
- This approach establishes a structured, repeatable foundation for Day 1 infrastructure onboarding and ongoing lifecycle operations.

AI SuperCluster

AI Supercluster module is purpose built for deploying, operating and scaling large GPU clusters as a single system, providing a unified infrastructure layer capable of supporting both AI and HPC infrastructure.

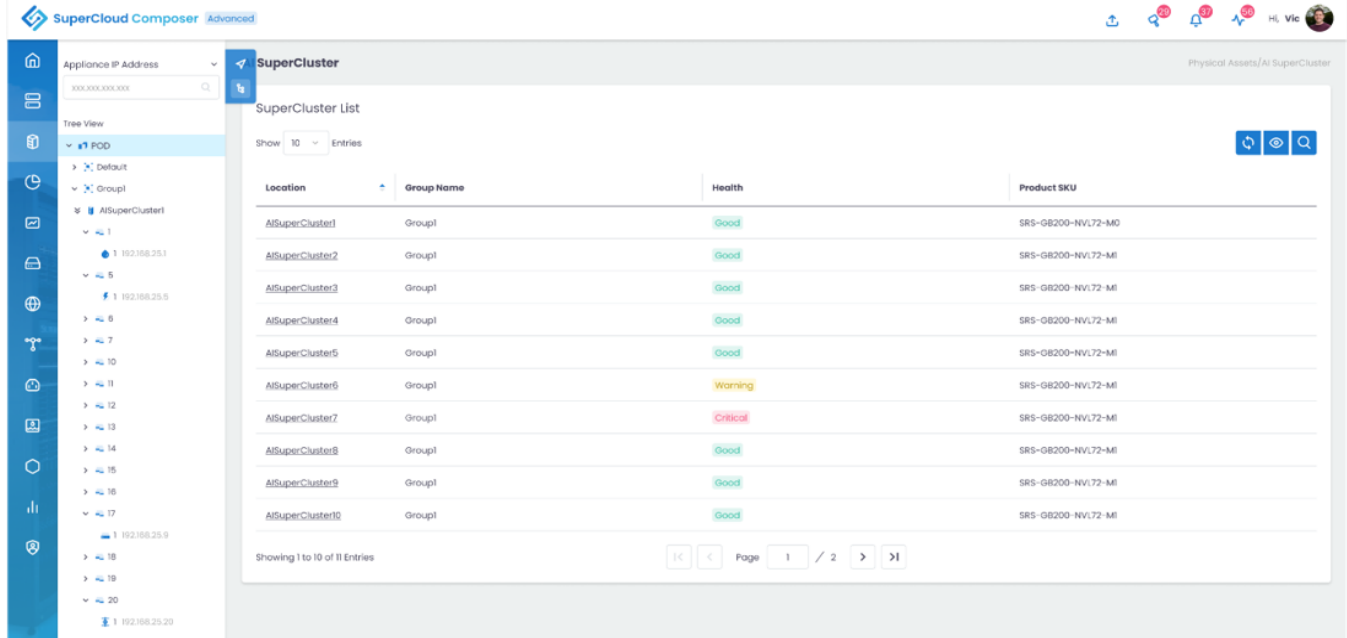


Figure 11. AI SuperCluster - List View

- Consistent orchestration across racks and SuperClusters, supporting coordinated deployment, scaling, and operations.
- Unified control of compute, network, fabric, power, and cooling, ensuring predictable performance and stability.
- Single operational view per cluster for managing high-density AI infrastructure.

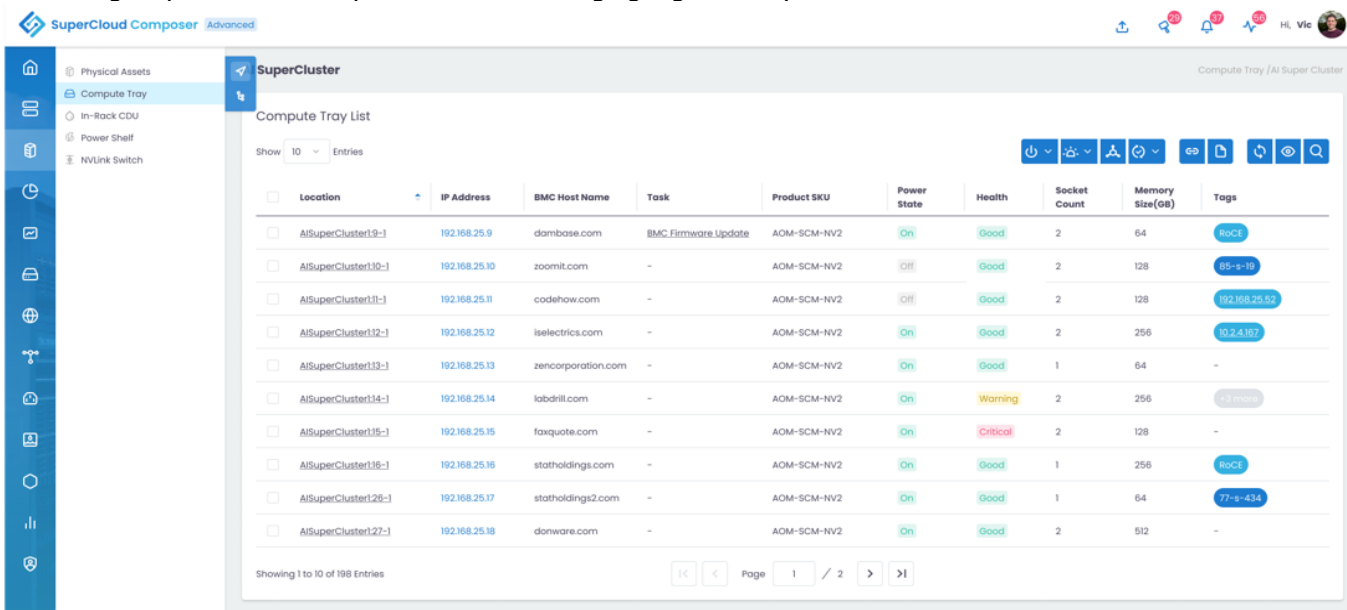


Figure 12. AI SuperCluster - Physical Assets

The AI SuperCluster module allows users to visualize all associated physical assets, including compute trays, CDUs, power shelves, and NVLink switches in a single view.

Location	IP Address	Product SKU	Power State	Health	State	Serial Number	Manufacturer	BMC Version
AI SuperCluster(20-1)	192.168.25.20	P3809	On	Good	Standby Offline	1583224800142	NVIDIA	00.00.03.07
AI SuperCluster(21-1)	192.168.25.21	P3809	Off	Good	Enable	1583224800143	NVIDIA	00.00.03.07
AI SuperCluster(22-1)	192.168.25.22	P3809	Off	Good	Disable	1583224800144	NVIDIA	00.00.03.07
AI SuperCluster(23-1)	192.168.25.23	P3809	On	Good	Standby Spare	1583224800145	NVIDIA	00.00.03.07
AI SuperCluster(24-1)	192.168.25.24	P3809	On	Good	In Test	1583224800146	NVIDIA	00.00.03.07
AI SuperCluster(25-1)	192.168.25.25	P3809	On	Warning	Starting	1583224800147	NVIDIA	00.00.03.07
AI SuperCluster(26-1)	192.168.25.26	P3809	On	Error	Absent	1583224800148	NVIDIA	00.00.03.07
AI SuperCluster(27-1)	192.168.25.27	P3809	On	Good	Unavailable Offline	1583224800149	NVIDIA	00.00.03.07
AI SuperCluster(28-1)	192.168.25.28	P3809	On	Good	Deferring	1583224800150	NVIDIA	00.00.03.07
AI SuperCluster(29-1)	192.168.25.29	P3809	On	Good	Quiesced	1583224800151	NVIDIA	00.00.03.07

Figure 13. AI SuperCluster – NVLink Details, Health & Operational State

- Supports full infrastructure lifecycle, from initial deployment through to production operation and incremental expansion, without re-design or re-integration.

Sensor Name	Status	Lower Critical	Lower Warning	Current Value	Upper Warning	Upper Critical
FAN 1 FRONT	Critical	2204 RPM	2500 RPM	2700 RPM	5000 RPM	6000 RPM
FAN 1 REAR	Critical	-	-	-	-	-
FAN 2 FRONT	Critical	2204 RPM	2500 RPM	2700 RPM	5000 RPM	6000 RPM
FAN 2 REAR	Critical	-	-	-	-	-
FAN 3 FRONT	Critical	2204 RPM	2500 RPM	2700 RPM	5000 RPM	6000 RPM
FAN 3 REAR	Critical	-	-	-	-	-
FAN 4 FRONT	Critical	2204 RPM	2500 RPM	2700 RPM	5000 RPM	6000 RPM
FAN 4 REAR	Critical	-	-	-	-	-
FAN 5 FRONT	Critical	2204 RPM	2500 RPM	2700 RPM	5000 RPM	6000 RPM

Figure 14. AI SuperCluster – NVLink Details, Health & Operational State

- Deep operational visibility and telemetry providing real-time insight into GPU health, thermal output, power consumption, and fabric status to enable proactive monitoring and faster fault isolation.

Monitoring Detail

Monitoring Details provides comprehensive, real-time and historical visibility across SuperCloud Composers managed infrastructure. It continuously collects health, performance, power, and thermal telemetry from systems spanning PODs, groups, racks, drawers, and individual devices, presenting this data through structured dashboards, charts, and reports. By correlating infrastructure state over time, the Monitoring Details module enables operators to assess system health, identify trends, and support informed operational and capacity decisions across enterprise, HPC, and AI environments.

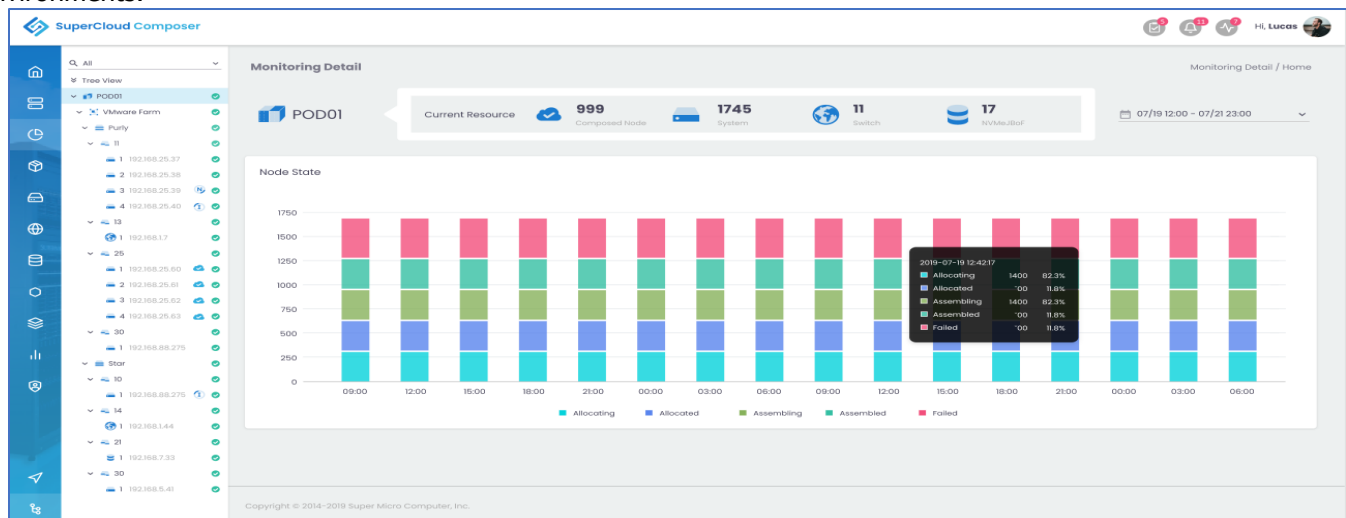


Figure 15. Monitoring Details – Pod View Utilization Overview

- Provides real-time monitoring and historical charting, across pods, racks, chassis and node levels.
- Analytics are presented through charts, graphs, and tables, providing at-a-glance visibility into performance metrics, telemetry, alerts, and historical trends.

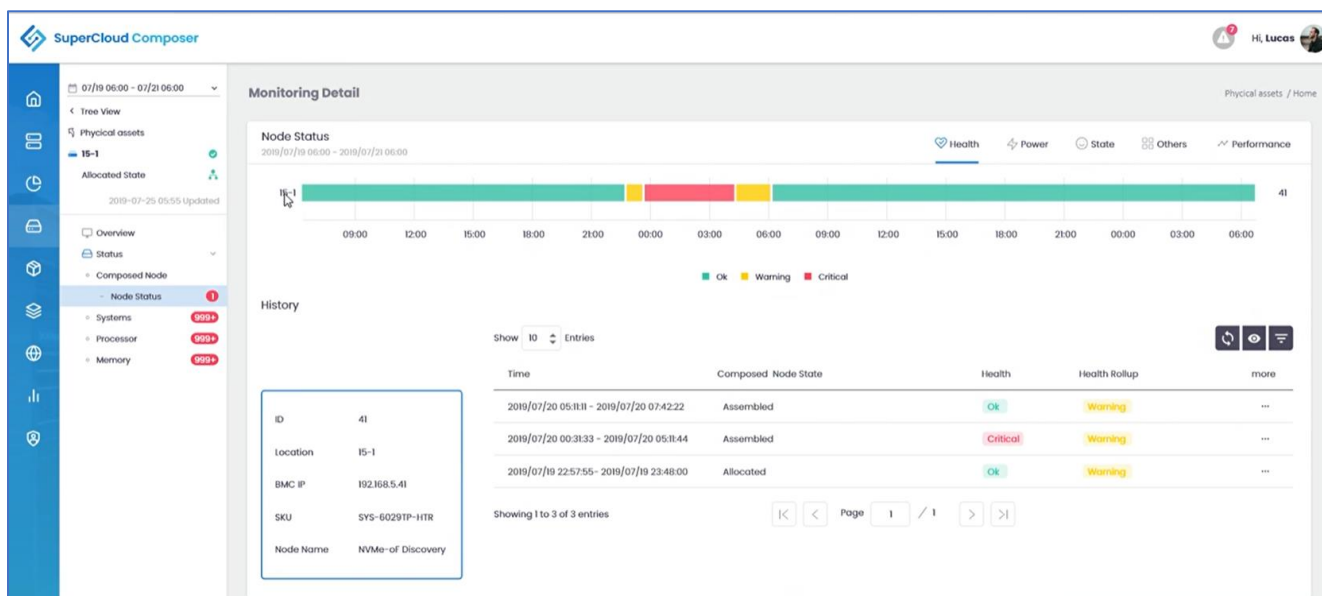


Figure 16. Health & Event Logs Collected - BMCs via Redfish

- SuperCloud Composer includes an analytics engine that collects and indexes real-time and historical operational data.
- Analytics services run on a Kubernetes-based backend, enabling scalable log collection and long-term trend analysis.
- Administrators can efficiently search and filter logs to support troubleshooting and scalable operations.

Performance Metrics

The Performance Metrics module delivers in-band telemetry for AI and HPC workloads, collecting component-level metrics from GPUs, Storage, TPUs, and DPUs.

Metrics are ingested via external Prometheus collectors and visualised through configurable, Grafana-style dashboards. Built-in charting enables clear visibility into utilisation, performance trends, and operational behavior of these resources, allowing for workload and threshold optimisations.

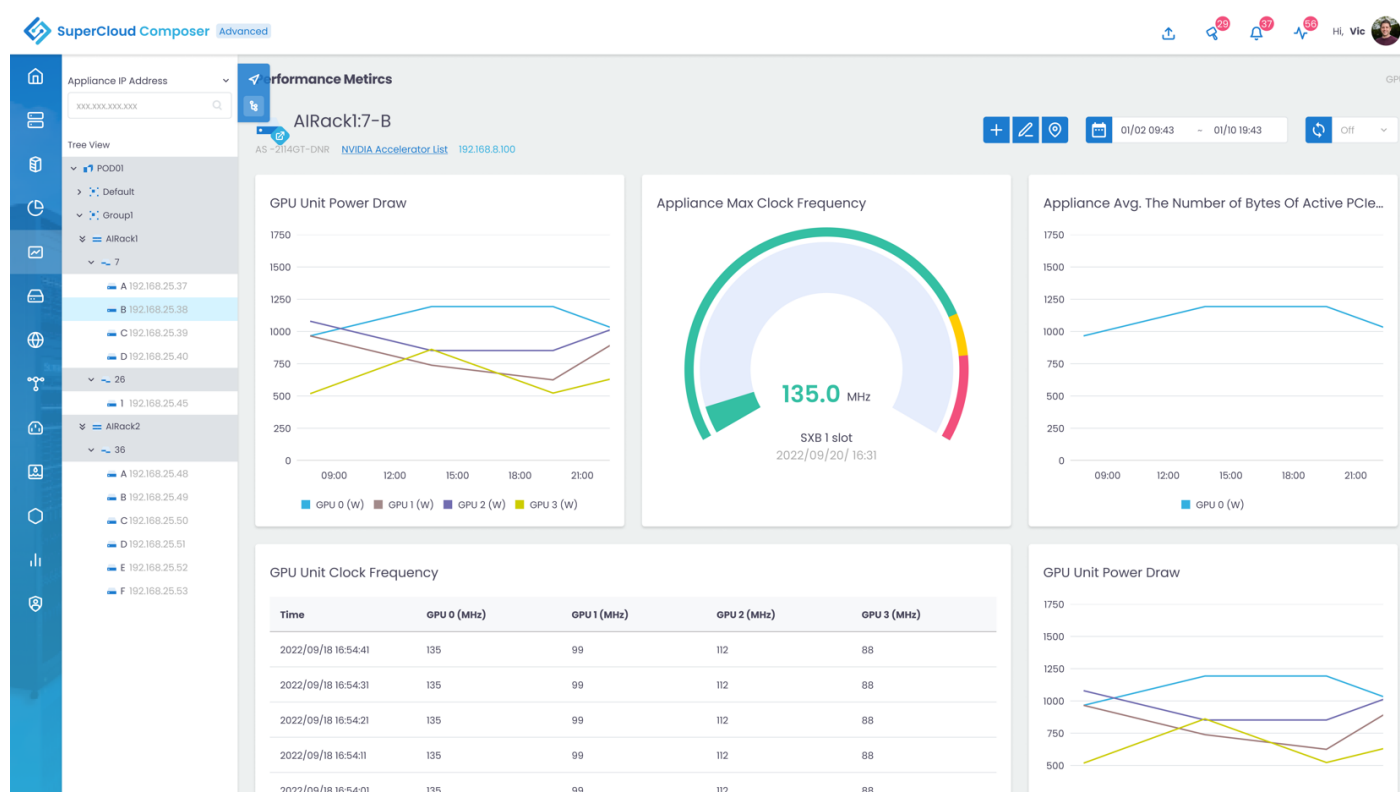


Figure 17. Real-Time GPU Utilization, Power & Clock Analytics

- Supports vendor specific GPUs telemetry exporters.
 - NVIDIA – via DCGM (Datacenter GPU Manager)
 - AMD – via ROCm (Radeon Open Compute platform)
 - Intel – via OneAPI (Intel's GPU telemetry framework)

The Performance Metrics module exposes in-band telemetry via Prometheus-compatible exporters, allowing customers to apply existing thresholding, alerting, and notification policies using their standard monitoring platforms.

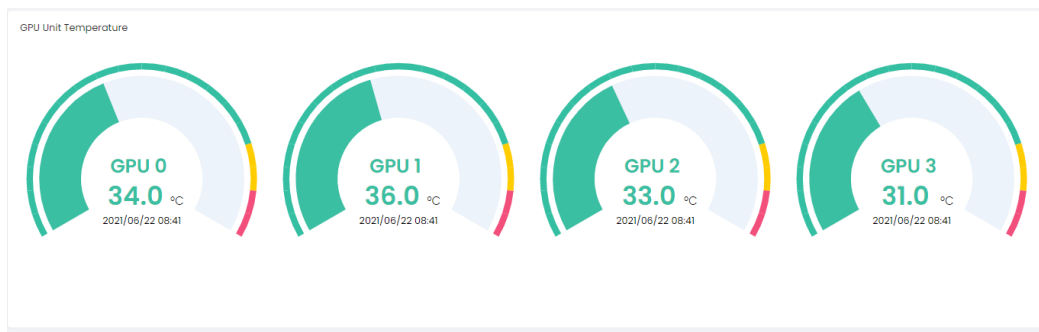


Figure 18. Per-GPU thermal telemetry (in-band), Live Temperature Readings & Threshold Alert Bands

- Customizable widgets, allowing for specific GPU related metadata visualization.
- Single Operational plane for Operators, Vendors and Engineers.

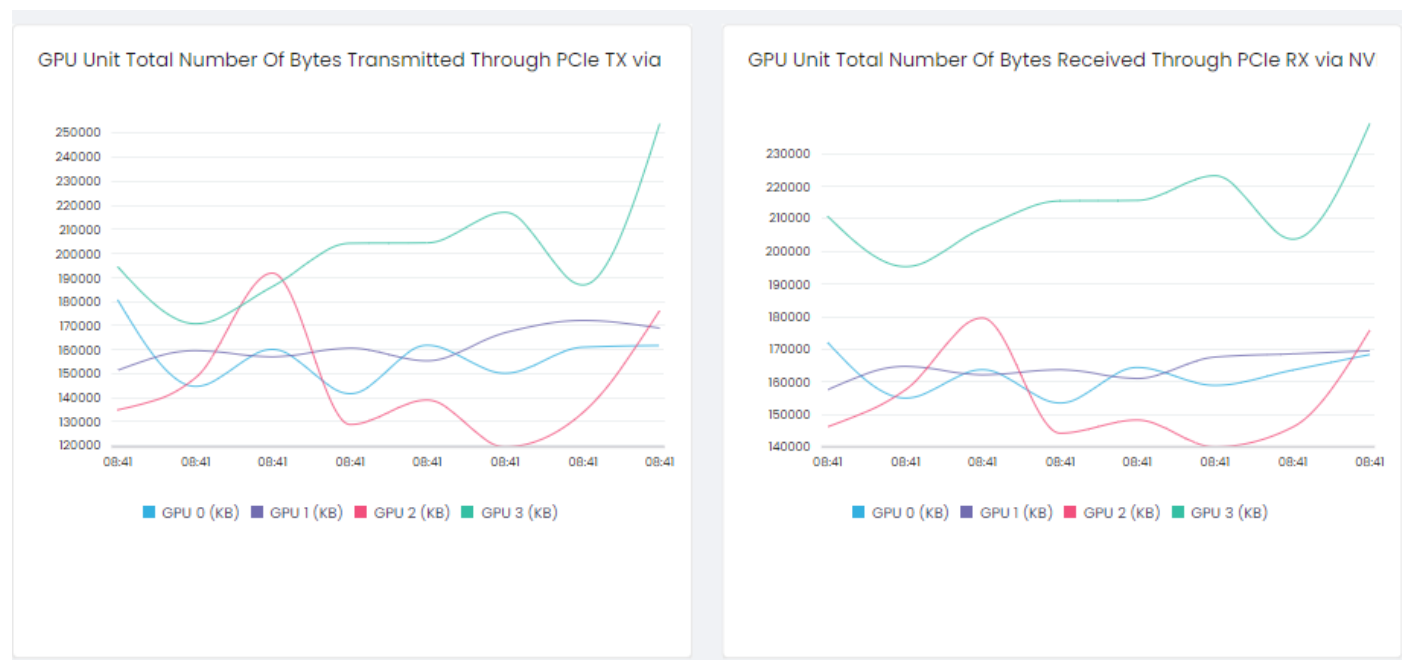


Figure 19. Per-Accelerator PCIe I/O Throughput Analysis

Compute Module

The Compute Module provides centralized lifecycle management for registered bare-metal computing resources across data centers. Enabling users to discover, provision, configure, and operate physical servers and accelerators in a consistent, repeatable manner, forming the foundation for higher-level platforms such as Kubernetes, schedulers, and virtualization stacks. A significant number of lifecycle management activities are actioned via the Compute Module.

The screenshot displays the 'Compute' section of the SuperCloud Composer interface. On the left is a sidebar with navigation icons and a tree view showing the hierarchy: P0001 > VMware Farm > Pury > 15 > 1 192.168.25.37, 2 192.168.25.38, 3 192.168.25.39, 4 192.168.25.40, 25 > 1 192.168.25.60, 2 192.168.25.61, 3 192.168.25.62, 4 192.168.25.63, 30 > 1 192.168.88.275, Star > 10 > 1 192.168.88.275, 30 > 1 192.168.5.41. The main area shows a 'Compute List' with 11 entries. The table columns are: Location, BMC IP, OS, Discovery State, Bios version, Bmc version, Power Status, Allocated, Health, CPU Cores, CPU Model, Memory Size, and Tags. The entries are filtered to show 1 to 10 of 11 entries. The bottom of the interface shows the copyright notice: Copyright © 2014-2019 Super Micro Computer, Inc.

Location	BMC IP	OS	Discovery State	Bios version	Bmc version	Power Status	Allocated	Health	CPU Cores	CPU Model	Memory Size	Tags
Pury15-1	192.168.7.162	Ubuntu 18	Deep	3.0a (01/12/2019)	1.64	On	Yes	Good	2	Intel(R) Xeon(R) Gold 6152 CPU @ 2.10GHz	366.2	SoCE, iSCSI: 10.2.0.181
Pury15-2	192.168.7.163	Ubuntu 18	Basic	2.1a (11/08/2018)	2.53	Off	No	Good	1	Intel(R) Xeon(R) CPU E5-2630 v3 @ 2.40GHz	399.7	SoCE
Pury15-3	192.168.7.164	Ubuntu 18	Deep	3.0a (01/12/2019)	3.42	On	Yes	Good	2	Intel(R) Xeon(R) Gold 6152 CPU @ 2.10GHz	433.2	SoCE
Pury15-4	192.168.7.165	Ubuntu 18	Deep	2.1a (11/08/2018)	4.31	Off	No	Good	2	Intel(R) Xeon(R) CPU E5-2630 v3 @ 2.40GHz	466.7	iSCSI: 10.2.0.12
Pury19-1	192.168.7.166	Ubuntu 18	Deep	3.0a (01/12/2019)	5.2	On	No	Good	2	Intel(R) Xeon(R) Gold 6152 CPU @ 2.10GHz	500.2	SoCE, iSCSI: 10.2.0.17
Pury19-2	192.168.7.167	Ubuntu 18	Deep	2.1a (11/08/2018)	6.09	Off	Yes	Good	1	Intel(R) Xeon(R) CPU E5-2630 v3 @ 2.40GHz	533.7	New
Pury25-1	192.168.7.168	Ubuntu 18	Deep	3.0a (01/12/2019)	6.98	On	Yes	Warning	2	Intel(R) Xeon(R) Gold 6152 CPU @ 2.10GHz	567.2	SoCE
Pury25-2	192.168.7.169	Ubuntu 18	Basic	2.1a (11/08/2018)	7.87	Off	No	Warning	2	Intel(R) Xeon(R) CPU E5-2630 v3 @ 2.40GHz	600.7	iSCSI: 10.2.0.25
Pury25-3	192.168.7.170	Ubuntu 18	Basic	3.0a (01/12/2019)	8.76	On	No	Warning	1	Intel(R) Xeon(R) Gold 6152 CPU @ 2.10GHz	634.2	SoCE, iSCSI: 10.2.0.12
Pury25-4	192.168.7.171	Ubuntu 18	Basic	2.1a (11/08/2018)	9.65	Off	No	Critical	2	Intel(R) Xeon(R) CPU E5-2630 v3 @ 2.40GHz	667.7	SoCE, Keep Crashing

Figure 20. Compute Module – Operations Toolbar, System Inventory, Tagging & Resource Summary

- Users can access common compute operations directly from the top toolbar, eliminating the need to navigate between modules.

The screenshot displays the 'System List' section of the SuperCloud Composer interface. The top toolbar contains icons for various operations: Refresh, Stop, Start, Restart, Power Off, Power On, Reboot, Clone, Delete, and Search. The table columns are: Location, Node Name, BMC IP, Task, Product Family, Product SKU, MotherBoard Type, Power Status, Health, Socket Count, Memory Size (GB), and Tags. The entries are filtered to show 1 to 10 of 11 entries. The bottom of the interface shows the copyright notice: Copyright © 2014-2019 Super Micro Computer, Inc.

Location	Node Name	BMC IP	Task	Product Family	Product SKU	MotherBoard Type	Power Status	Health	Socket Count	Memory Size (GB)	Tags
Compute Rack3-C	-	172.24.169.40	-	Twin Family	SYS-2029BT-HNCOR	X11DPT-B+	On	Good	2	384	RoCE, NONE
Compute Rack3-D	-	172.24.169.41	-	Twin Family	SYS-2029BT-HNCOR	X11DPT-B+	On	Good	2	384	RoCE, NONE
Compute Rack9-B	-	172.24.169.35	-	Twin Family	AS-2115GT-HNTR	H13SST-G	On	Critical	1	288	-
Compute Rack12-A	-	172.24.169.32	-	Twin Family	SYS-220TP-HTTR	X12DPT-PT6/PT46	On	Critical	2	64	101
Compute Rack12-B	-	172.24.169.42	-	Twin Family	SYS-220TP-HTTR	X12DPT-PT6/PT46	On	Critical	2	64	-
Compute Rack12-C	-	172.24.169.33	-	Twin Family	SYS-220TP-HTTR	X12DPT-PT6/PT46	On	Critical	2	64	-
Compute Rack12-D	-	172.24.169.34	-	Twin Family	SYS-220TP-HTTR	X12DPT-PT6/PT46	On	Critical	2	64	111
Compute Rack14-A	-	172.24.169.59	-	Twin Family	SYS-620BT-DNCBR	X12DPT-B	On	Good	2	320	Supermicro, as

Figure 21. Compute Module – Operations Toolbar Filtering Across Fleets

- Filter and search across all compute asset metadata.
- Tag Management – Tags can be assigned to systems to assist in filtering and automation activities.

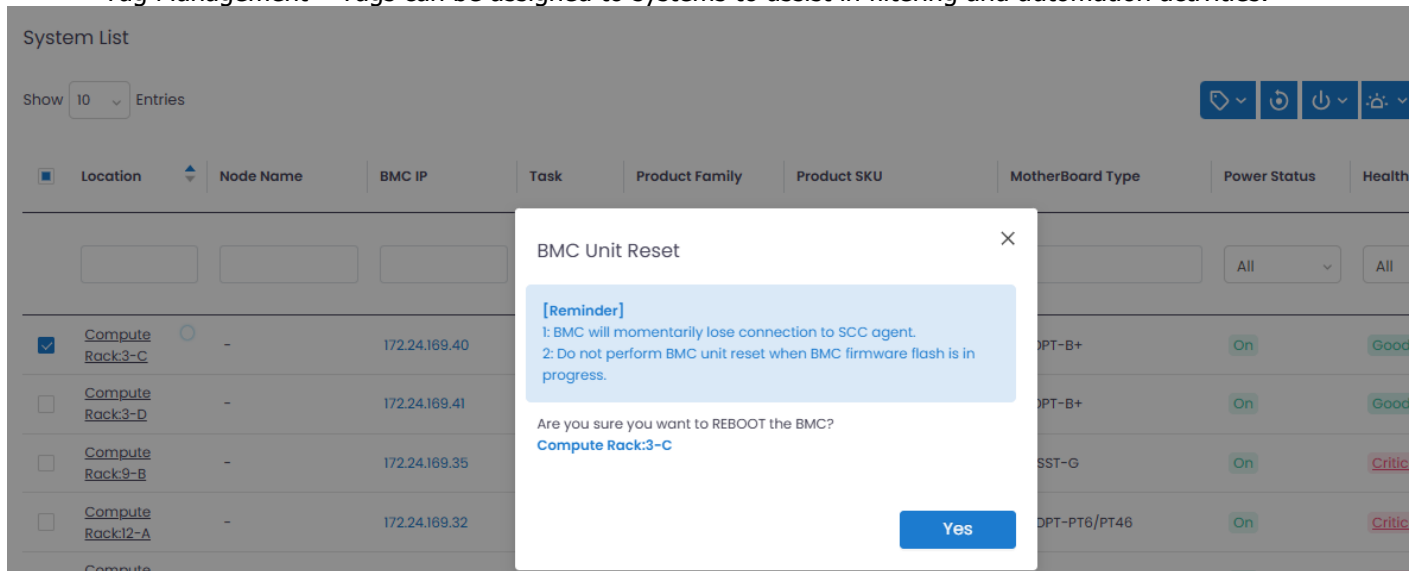


Figure 22. Compute Module – BMC Unit Reset

- Remote Power Actions – Power On, Force Off, Force Restart and Graceful Shutdown power actions.

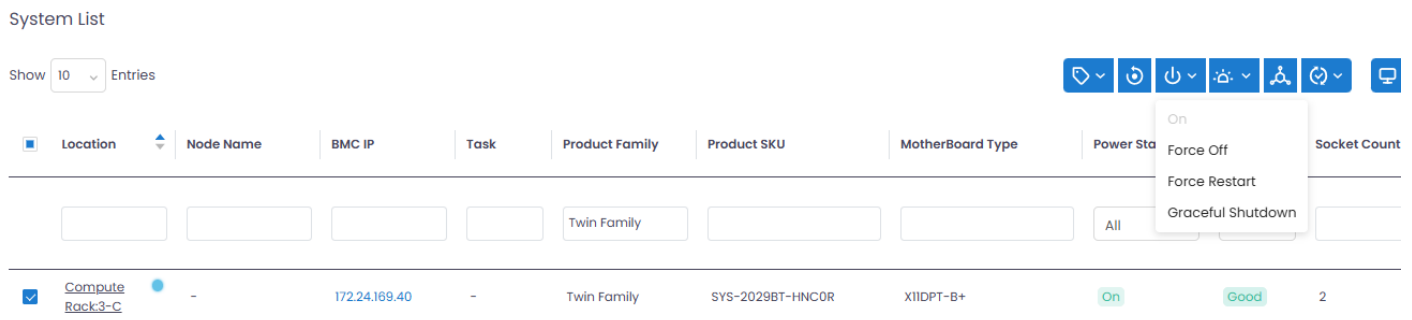


Figure 23. Compute Module – System Power Actions

- UID Light – Remotely turn on/off the unit identifier blinker to identify assets faster in densely populated racks.

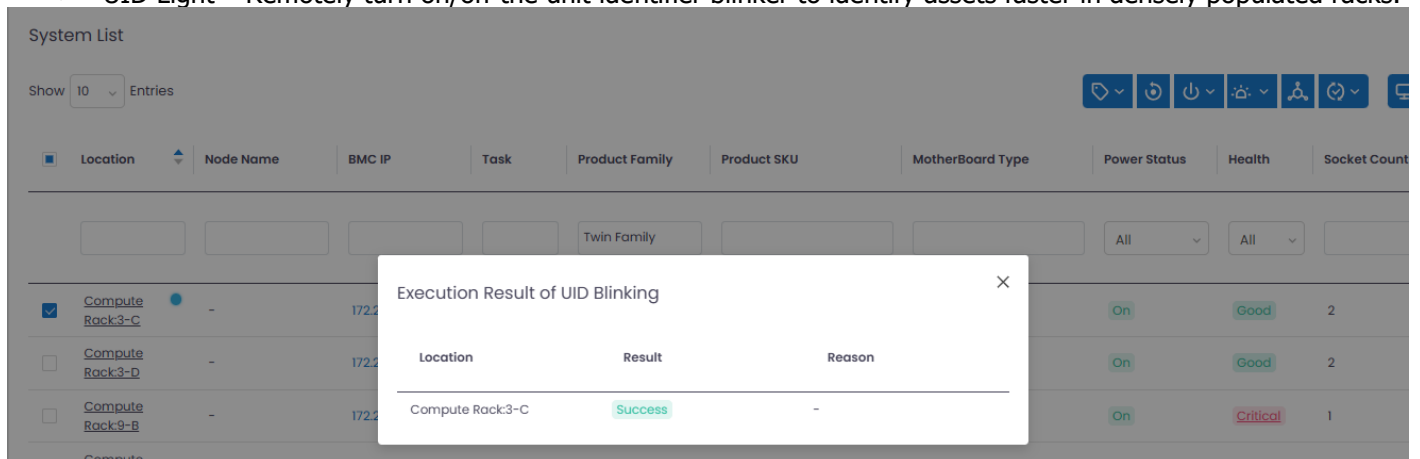


Figure 24. Compute Module – UID Identifier

- Deploy OS – Image-based bare-metal OS provisioning using out of band.

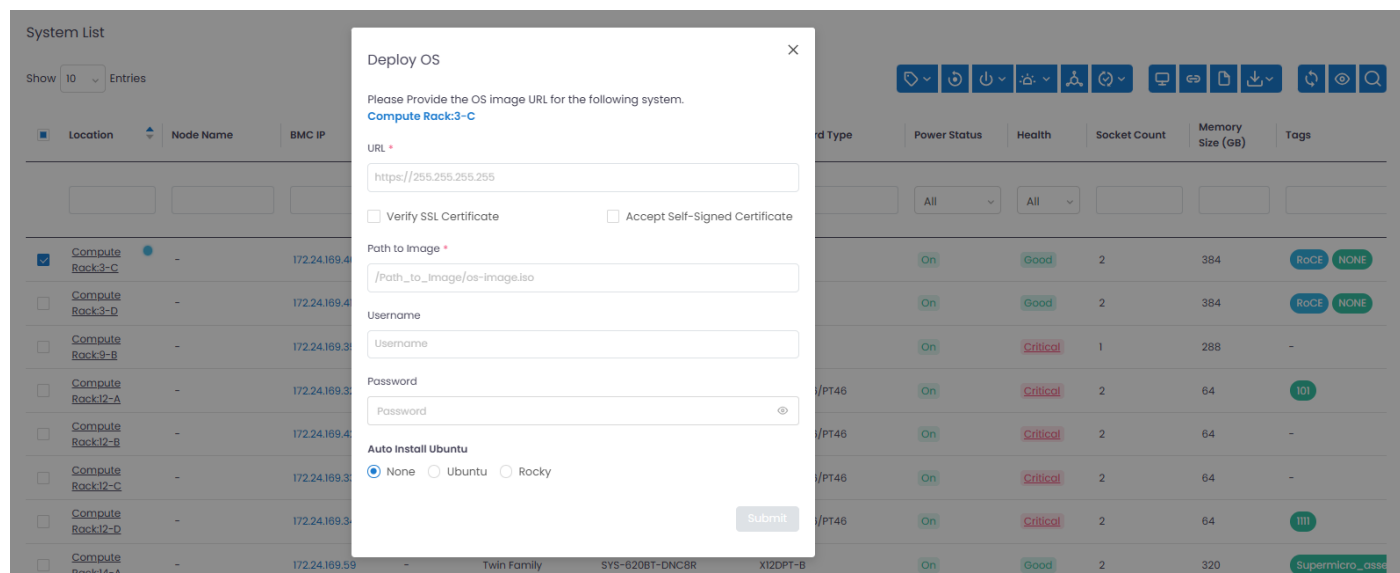


Figure 25. Compute Module – OS Deployment

- Firmware – BIOS and BMC firmware updates via secure local repositories.

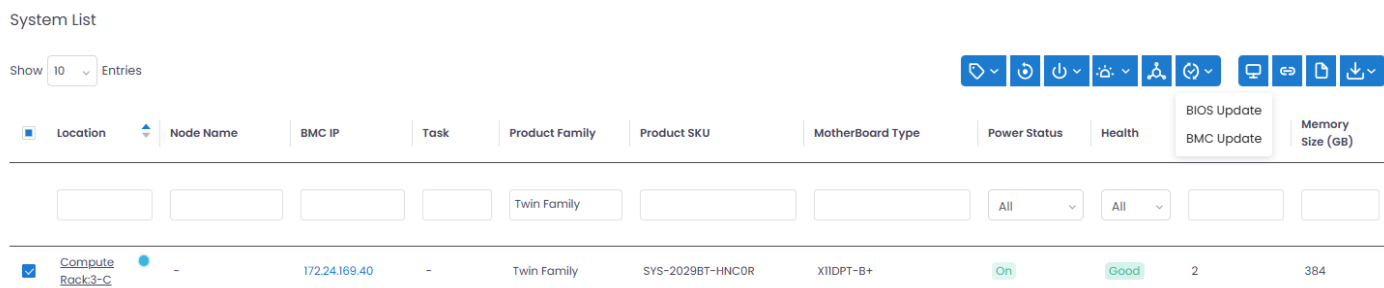


Figure 26. Compute Module – BIOS & BMC Update

- BMC IP - Access to a systems BMC via out of band management IP.

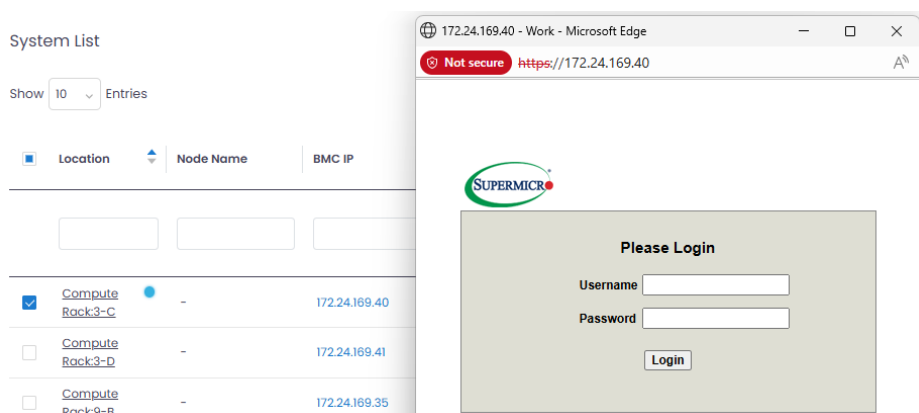


Figure 27. Compute Module – One Click to Systems BMC

- Physical Assets – Provides a detailed view of compute system layout and hardware metadata, including location, manufacturer, serial number, power and system state, FRU information, and additional component-level details.

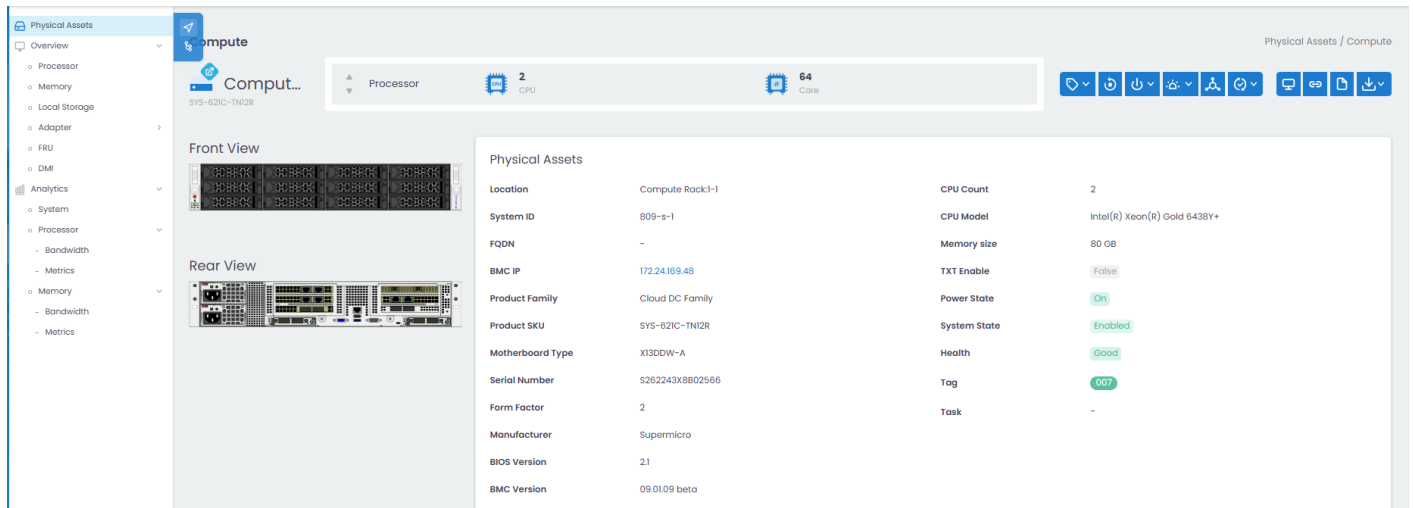


Figure 28. Compute Module – System Details & State

- Detailed visibility of internal system components, exposing processor, memory, accelerator, and hardware health information.

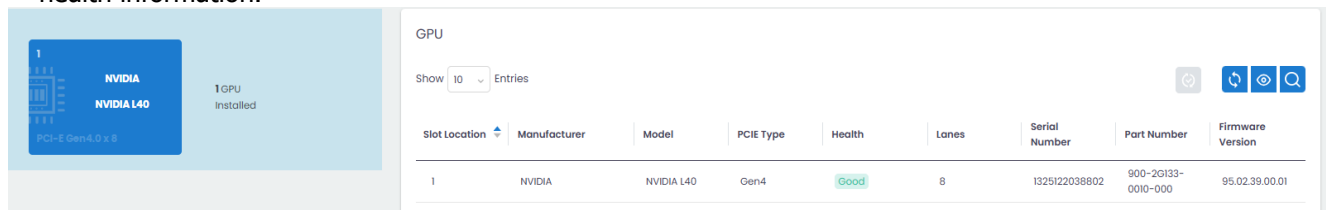


Figure 29. Compute Module – Physical Assets GPU Details & State

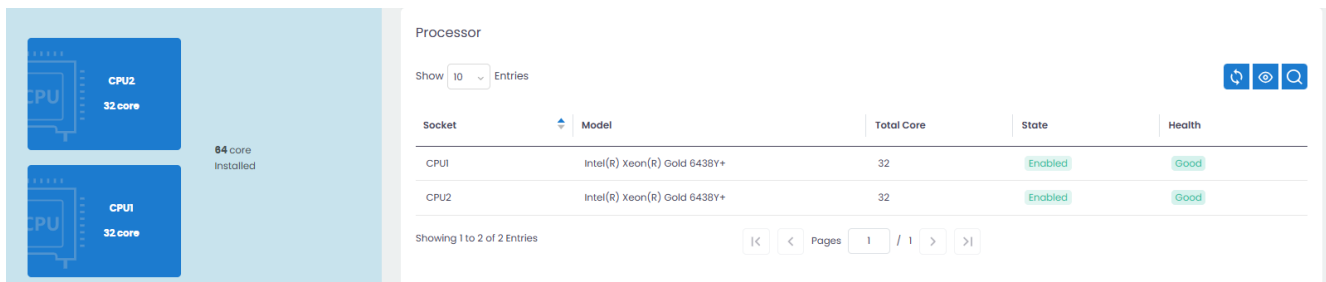


Figure 30. Compute Module – Physical Assets CPU Details & State

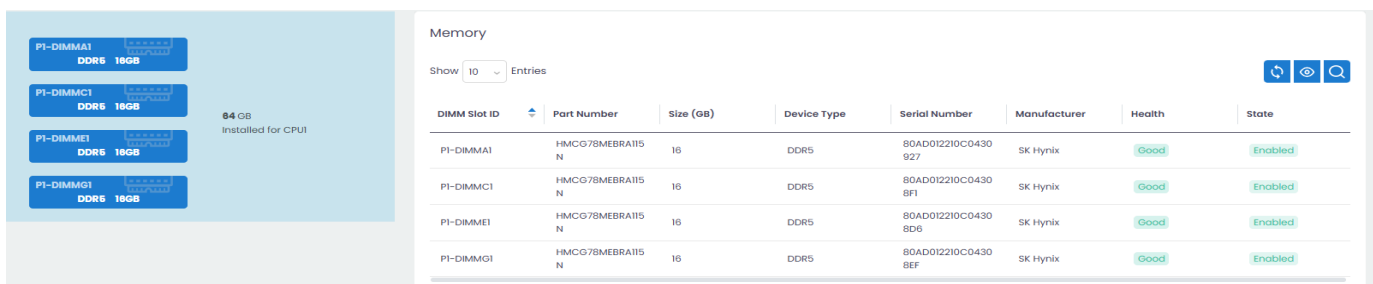


Figure 31. Compute Module – Physical Assets Memory Details & State

- Advanced time-series visual analytics for CPU and memory metrics, delivered through configurable dashboards with historical export.

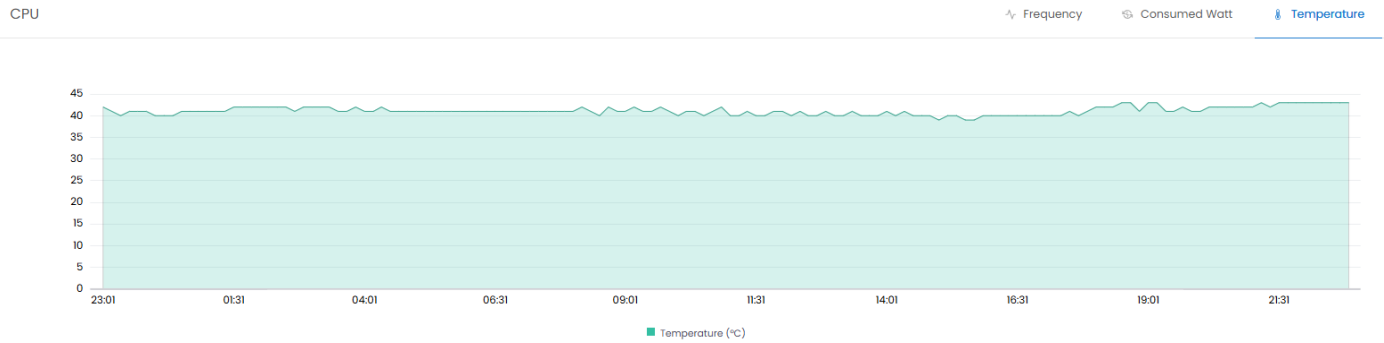


Figure 32. Compute Module – Physical Assets CPU Telemetry (Temperature)

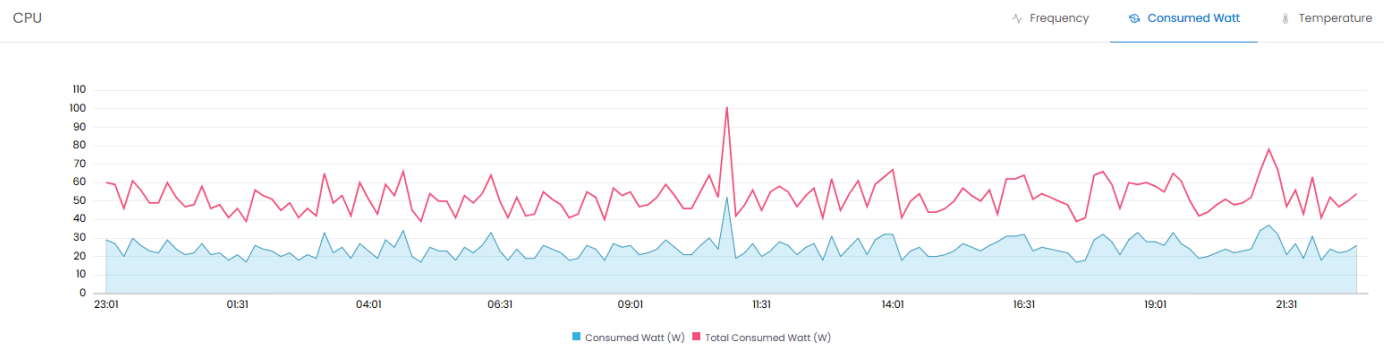


Figure 33. Compute Module – Physical Assets CPU Telemetry (Consumed Watt)

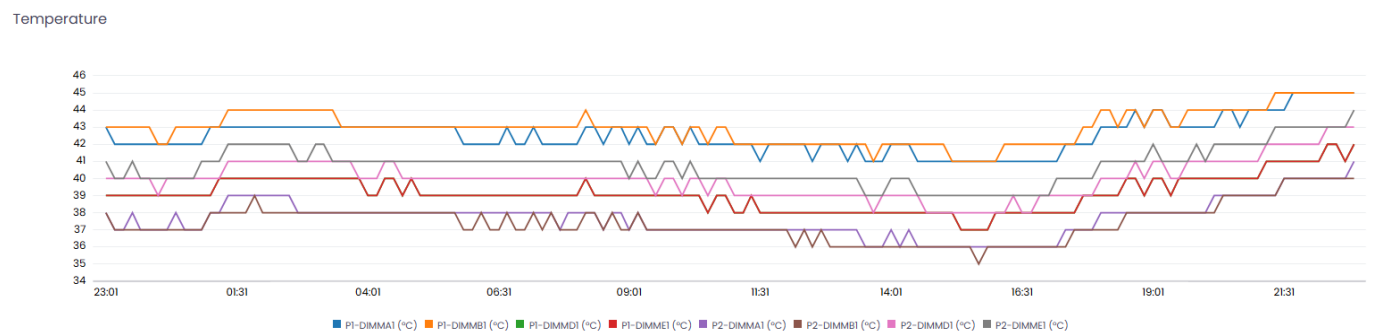


Figure 34. Compute Module – Physical Assets Memory Telemetry (Temperature)

Network Module

The Network module provides centralized provisioning, orchestration, and compliance management for data center switching infrastructure.

Physical Assets

Uplink Port Server Port Fabric Port

Switch ID	Port Number	OP. Status	Type	Speed	Duplex	Negotiation	VLANs Allowed	VLAN Mode	Native VLAN	Flow Control	LAG	Raw Data
50	Cx0/1	Up	Uplink	100 GBPS	Full	Auto	[1, 2, 3, 4050]	Trunk	1	TRUE	po25	
52	Cx0/2	Up	Uplink	101 GBPS	Full	Auto	[1, 2, 4, 4050]	Trunk	1	TRUE	po26	
54	Cx0/3	Down	Server	102 GBPS	Full	Auto	[1, 2, 5, 4050]	Trunk	1	TRUE	po27	
56	Cx0/4	Up	Server	103 GBPS	Full	Auto	[1, 2, 6, 4050]	Trunk	1	TRUE	po28	
58	Cx0/5	Down	-	104 GBPS	Full	Auto	[1, 2, 7, 4050]	Trunk	1	TRUE	po29	
60	Cx0/6	Down	-	105 GBPS	Full	Auto	[1, 2, 8, 4050]	Trunk	1	TRUE	po30	
62	Cx0/29	Up	Fabric	106 GBPS	Full	Auto	[1, 2, 9, 4050]	Trunk	1	TRUE	po31	
64	Cx0/30	Down	Fabric	107 GBPS	Full	Auto	[1, 2, 10, 4050]	Trunk	1	TRUE	po32	
66	Fx0/1	Up	Uplink	108 GBPS	Full	Auto	[1, 2, 11, 4050]	Trunk	1	TRUE	po33	
68	Fx0/2	Up	Uplink	109 GBPS	Full	Auto	[1, 2, 12, 4050]	Trunk	1	TRUE	po34	

Showing 1 to 3 of 3 entries

Figure 35. Network Module – Physical Assets Switch View

- SuperCloud Composer provides automated network provisioning, executing per-switch build plans sequentially or in parallel to apply consistent network intent across multiple data switches.
- Build plans are generated through the Network Configurator as JSON-based configurations and executed by the Network Orchestrator using standards-based switch APIs.
- During ongoing operations, Switch Sweeper validates live switch configurations against the intended build plans to detect and enforce configuration compliance

Network Module - Storage Fabric

The Storage Fabric component of SuperCloud Composer enables automated provisioning and compliance management for Ethernet-based storage fabrics used in bare-metal environments.

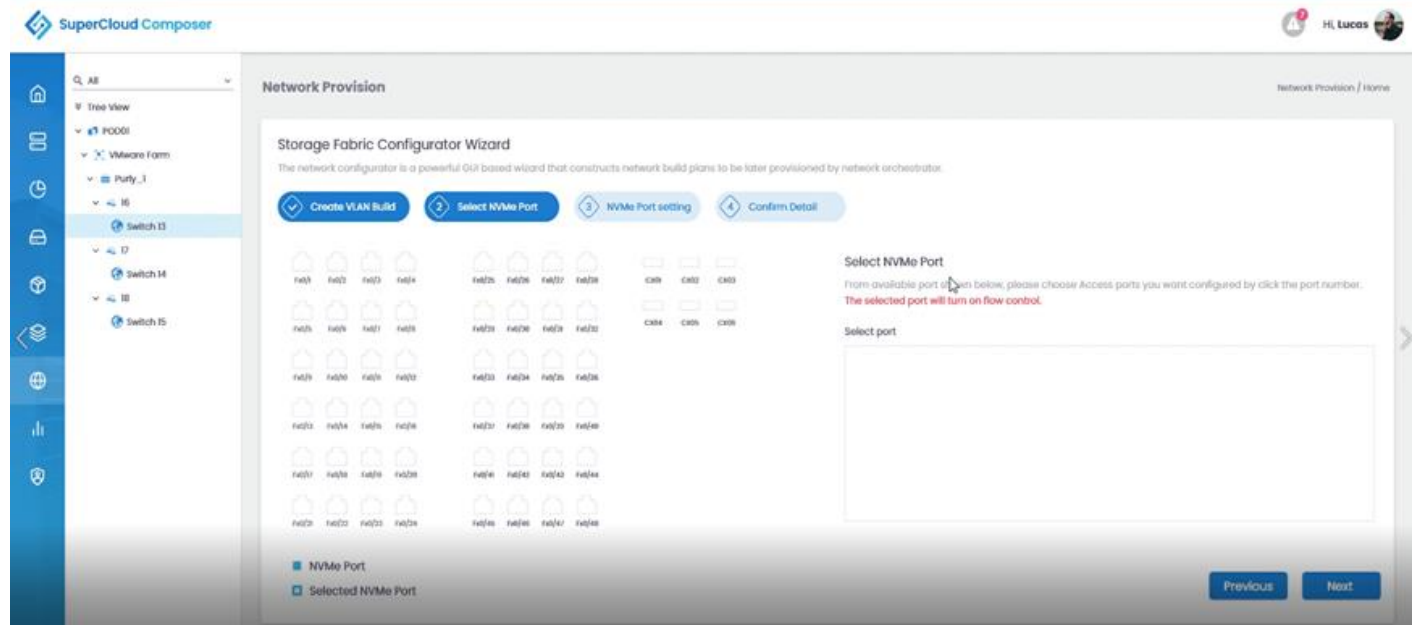


Figure 36. Network Module – Storage Fabric View

- Storage fabric provisioning applies to physical (bare-metal) servers only and requires RDMA/RoCE-capable NICs to enable NVMe-oF (NVMe over Fabrics) communication over Ethernet. (This capability does not apply to virtualized server networking.)
- Unlike general-purpose compute networking, storage fabrics require additional IEEE Data Center Bridging Exchange (DCBX) configuration to support lossless Ethernet transport for NVMe-oF workloads.
- RDMA/RoCE constructs rely on ETS, flow control, and lossless queues to maintain robust communication between NVMe OF targets, initiators, and Ethernet-based switches.
- Storage switch build plan templates are constructed by a Storage Fabric Configurator wizard formatted in JSON and pushed by a Network Orchestrator engine utilizing industry standardized API calls.
- During network management operation SuperCloud Composer offers a rich, intelligent network agent called switch sweeper to maintain configuration compliance between original build plans constructed by storage fabric configurator and operational build plans that exist within switch dynamic memory.

Topology Module

The Topology View provides a visual representation of physical connectivity between servers and management, storage, and compute switches at rack and drawer levels, offering clear visibility into how systems are interconnected.

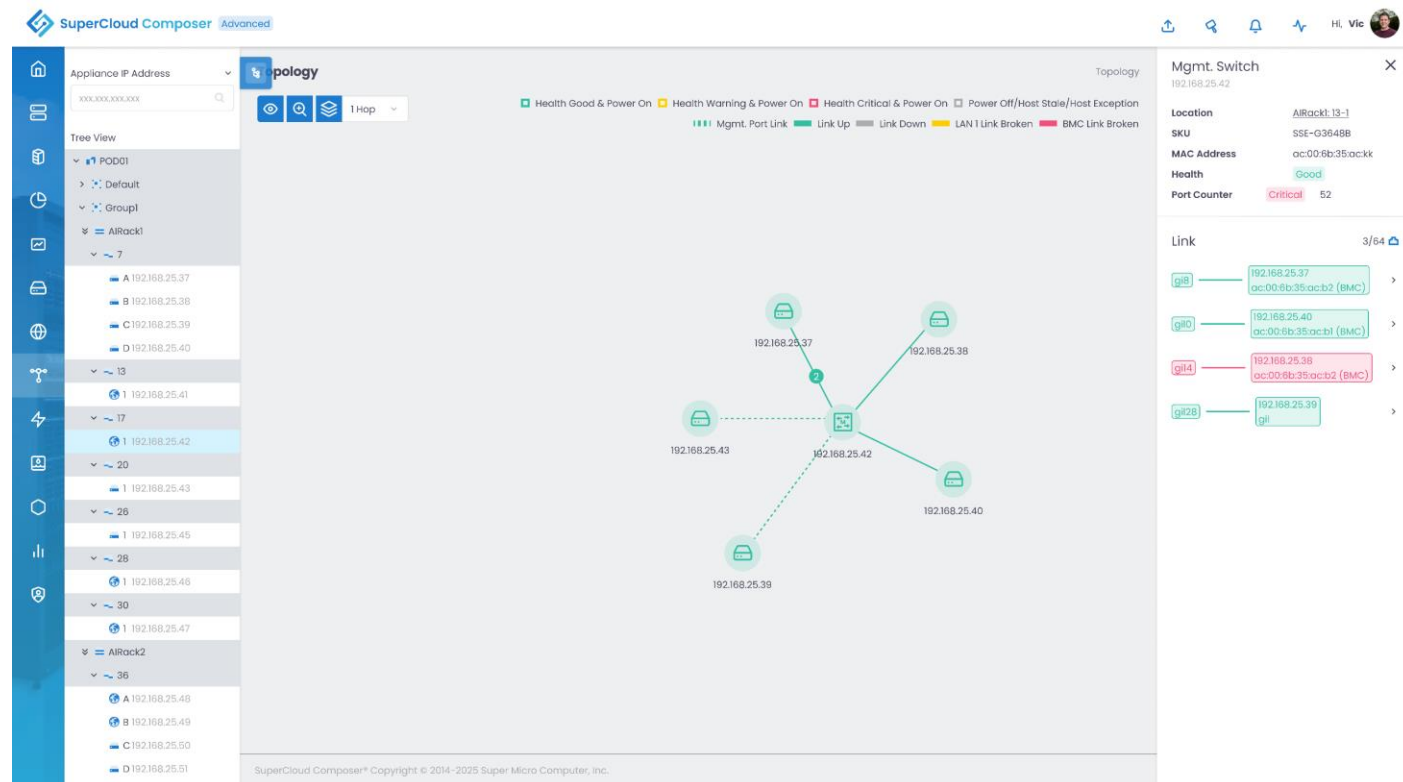


Figure 37. Topology Module – 1 Hop View

- Visualizes physical connectivity between servers and management, compute, and storage switches.
- Displays real-time link status using clear visual indicators for link up/down, warnings, and critical conditions.
- Provides health and power state visibility for connected systems directly within the topology view.
- Supports hop-based depth view of connectivity.
- Enables hierarchical navigation through racks, drawers, and systems via the tree view.
- Surfaces device-level details on selection, including switch or server metadata, port status, and MAC addressing.
- Highlights faults and exceptions (broken links, host power-off states) for fast identification.
- Offers a clear, intuitive map of network relationships, helping users quickly understand how infrastructure is wired and interconnected.

PDU Console

PDU Console delivers centralized, multi-vendor power visibility and control, combining outlet-level management, power analytics, and environmental monitoring within the SuperCloud Composer platform. Allowing users to define server-to-outlet power mappings using intelligent PDUs, with support for cascaded PDU configurations, integrated to support Schneider Electric (APC), Legrand and Schleifenbauer across POD and Rack scopes

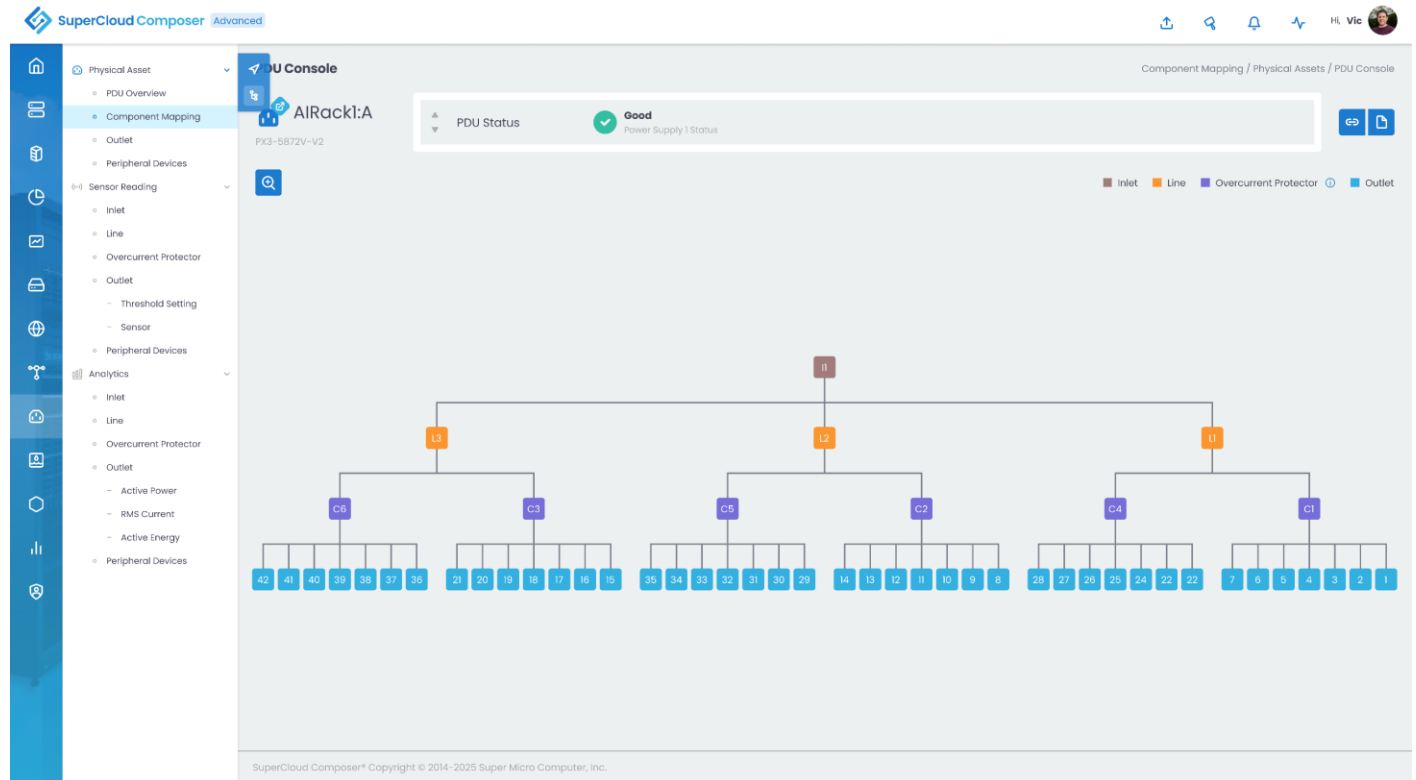


Figure 38. PDU Console – Power Distribution Mapping (PDU, Circuits, Outlets)

- Centralized PDU Inventory & Visibility - Unified view of rack-level and outlet-level power status, load, and consumption.
- Server-to-outlet power mapping - Maps physical servers to PDU outlets, enabling clear visibility of power dependencies and impact.

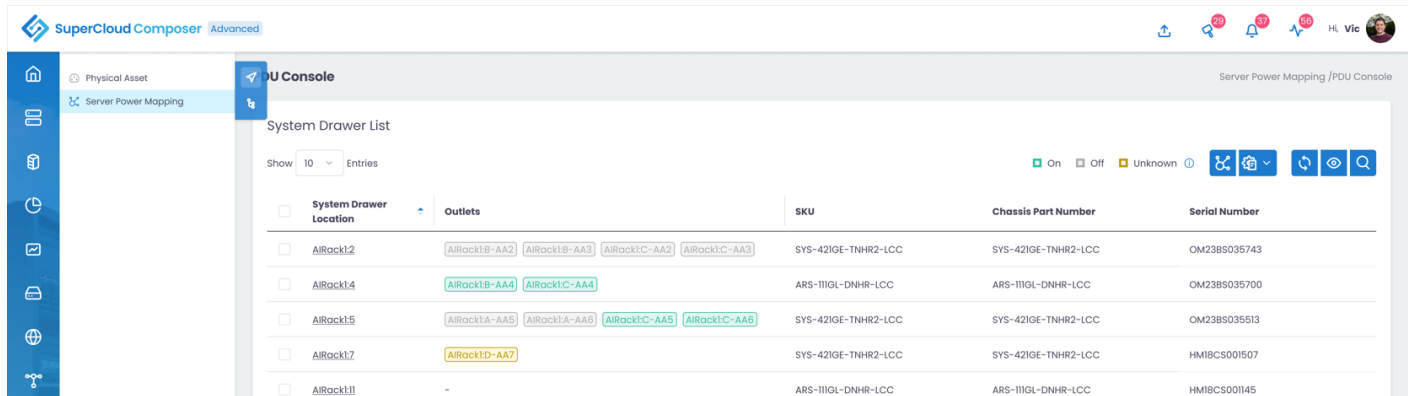


Figure 39. PDU Console – Server To PDU Power Mapping

- Outlet-Level Monitoring & Control - Remote on/off/cycle operations and per-outlet monitoring (vendor-dependent).
- Integrated power and environmental analytics, power, energy, temperature, and humidity metrics with threshold-based alerting.

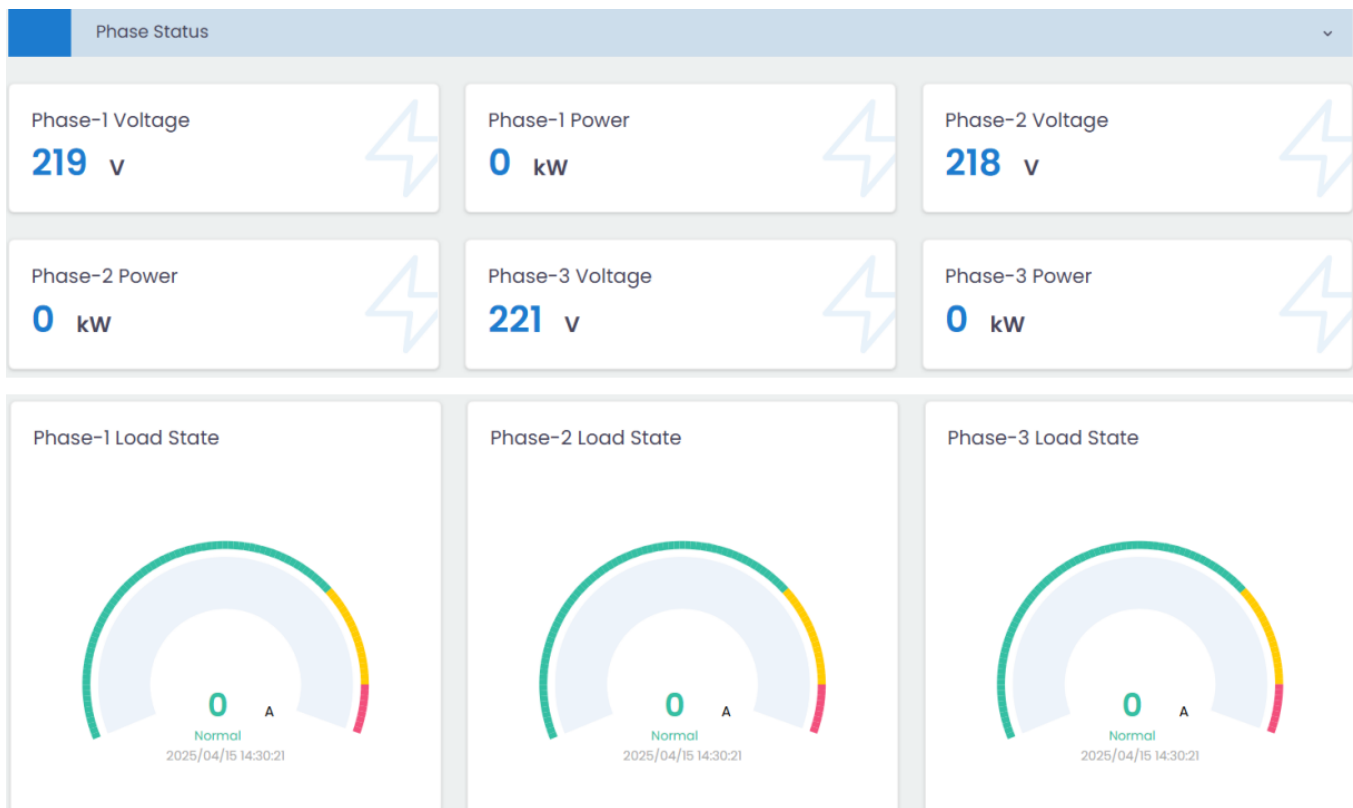


Figure 40. PDU Console – Three-Phase Power Monitoring Dashboard

- Visual telemetry dashboards, Interactive time-series charts for power and environmental data.

Bank-1 Outlet Energy

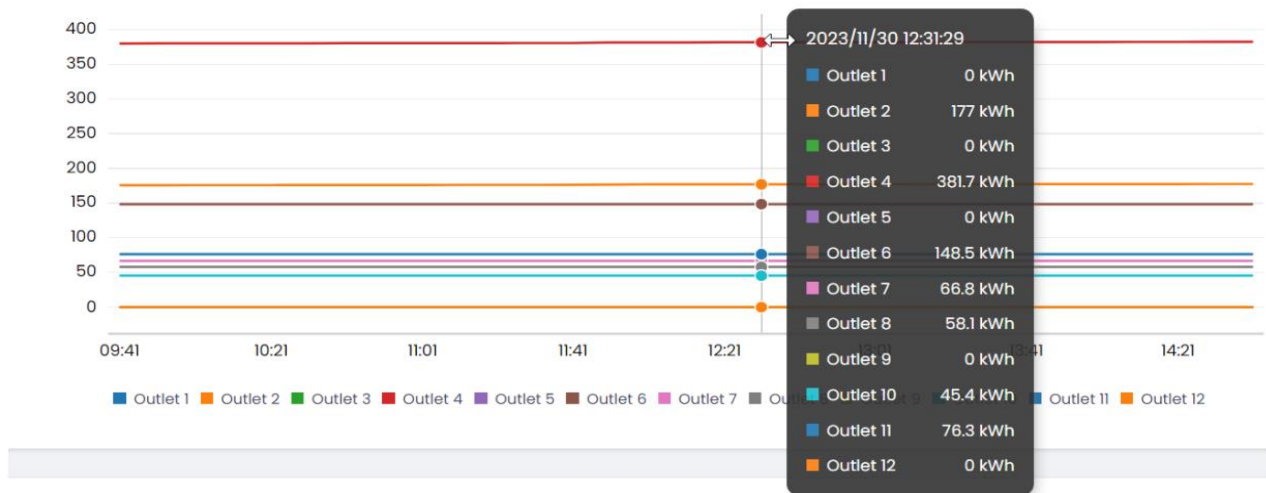


Figure 41. PDU Console – Granular Outlet-Level Energy Usage Analytics (kWh), Time-Series

Cooling Infrastructure

The Cooling Infrastructure module delivers unified visibility, telemetry, analytics, and control across liquid and facility cooling systems, enabling operators to monitor, analyze, and manage cooling performance at scale.

Supercloud composer supports a variety of different cooling architectures.

Cooling Tower - Facility-level cooling assets integrated into SCC to provide visibility into upstream cooling infrastructure and environmental telemetry.

- Physical asset inventory including firmware, SKU, serial number, manufacturer, location, and operational state.

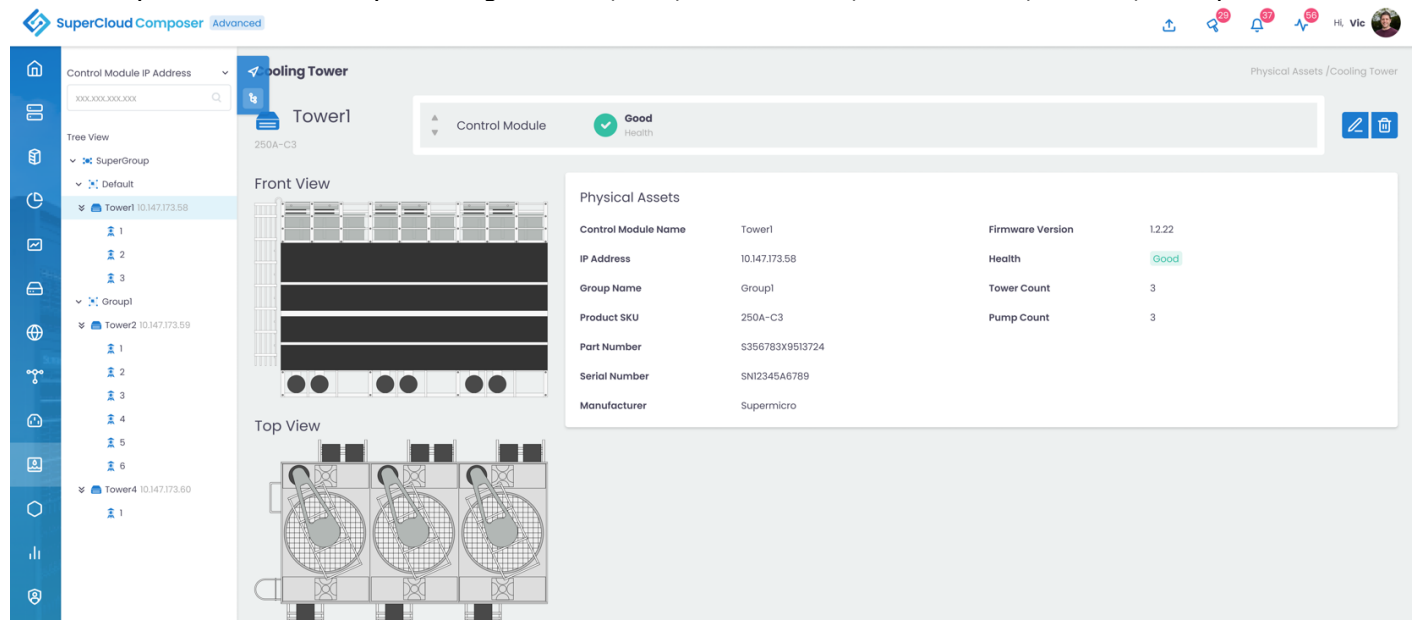


Figure 42. Cooling Infrastructure – Cooling Tower Details & State

In-Rack resident liquid cooling controllers delivering coolant directly to liquid-cooled servers, with real-time monitoring of pumps, valves, pressure, flow rate, temperatures, humidity, dew point, and heat load.

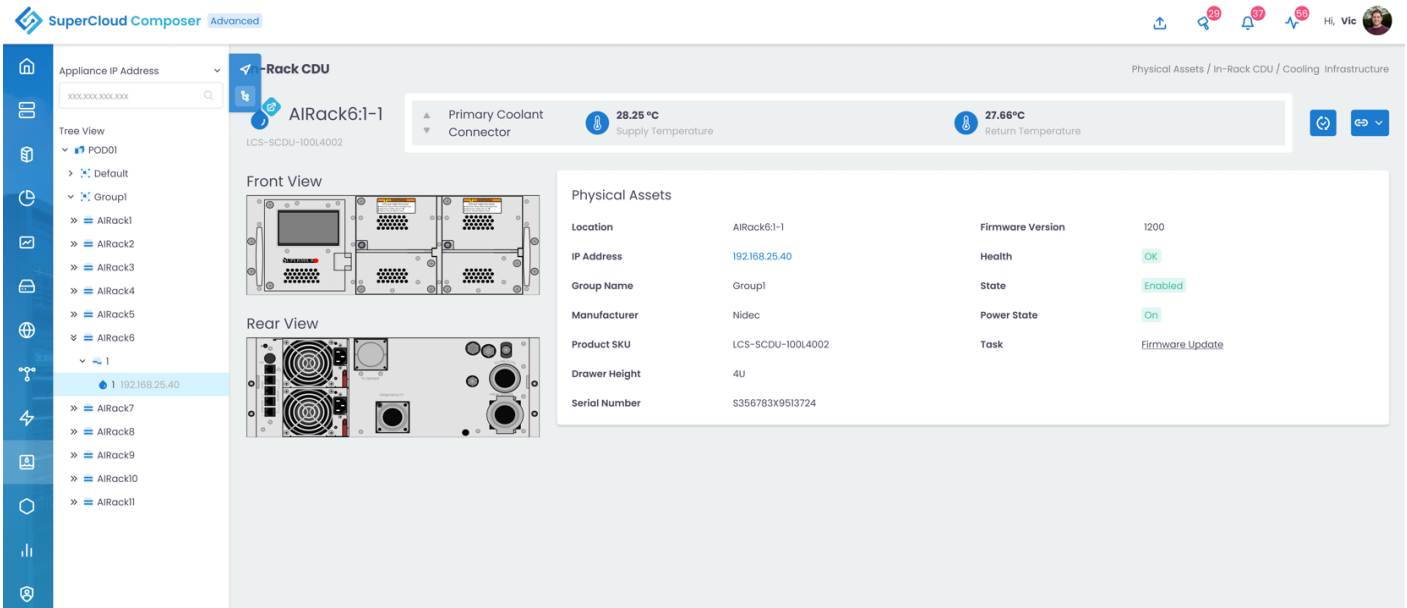


Figure 43. Cooling Infrastructure – In-Rack CDU Details & State

In-Row liquid cooling infrastructure delivers centralized coolant distribution across multiple racks, with integrated monitoring of coolant quality, pressure, temperature, and consumption.

- Hierarchical visibility across POD, group, row, rack, and device levels, aligned to physical data-center layout.

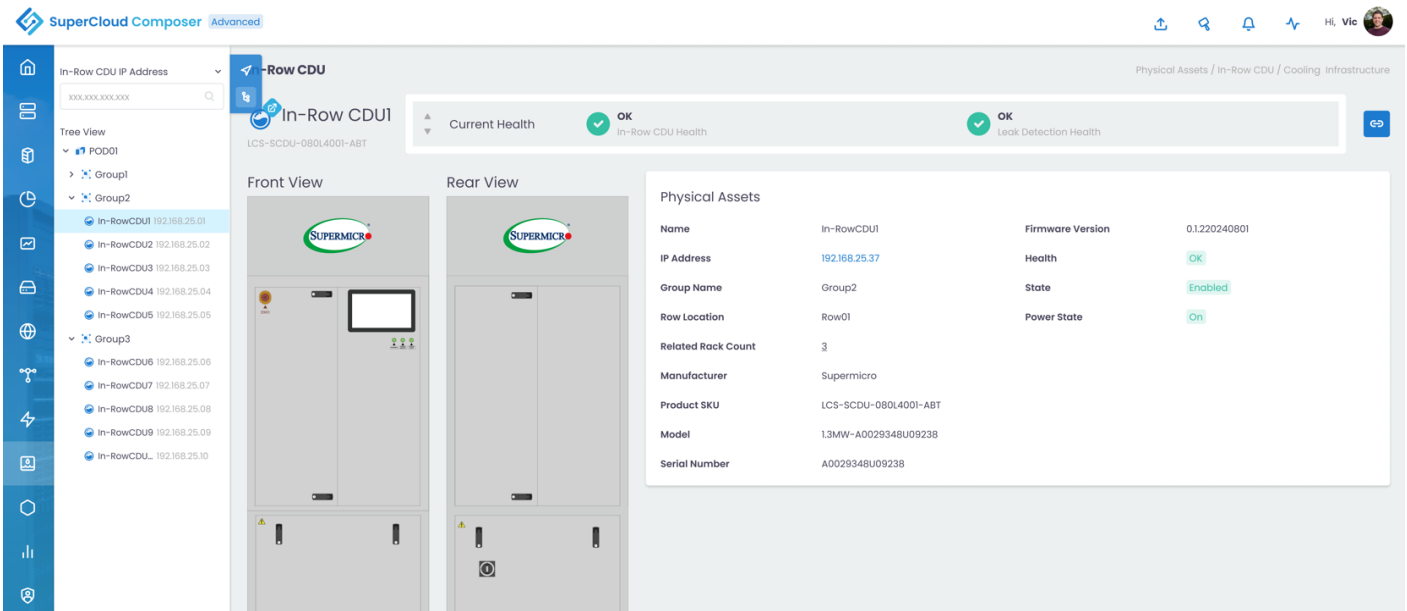


Figure 44. Cooling Infrastructure – In-Row CDU Details & State

Adjacent sidecar liquid cooling cabinets support liquid-to-air and liquid-to-liquid heat rejection for high-density deployments, with monitoring of power, temperature, and energy consumption.

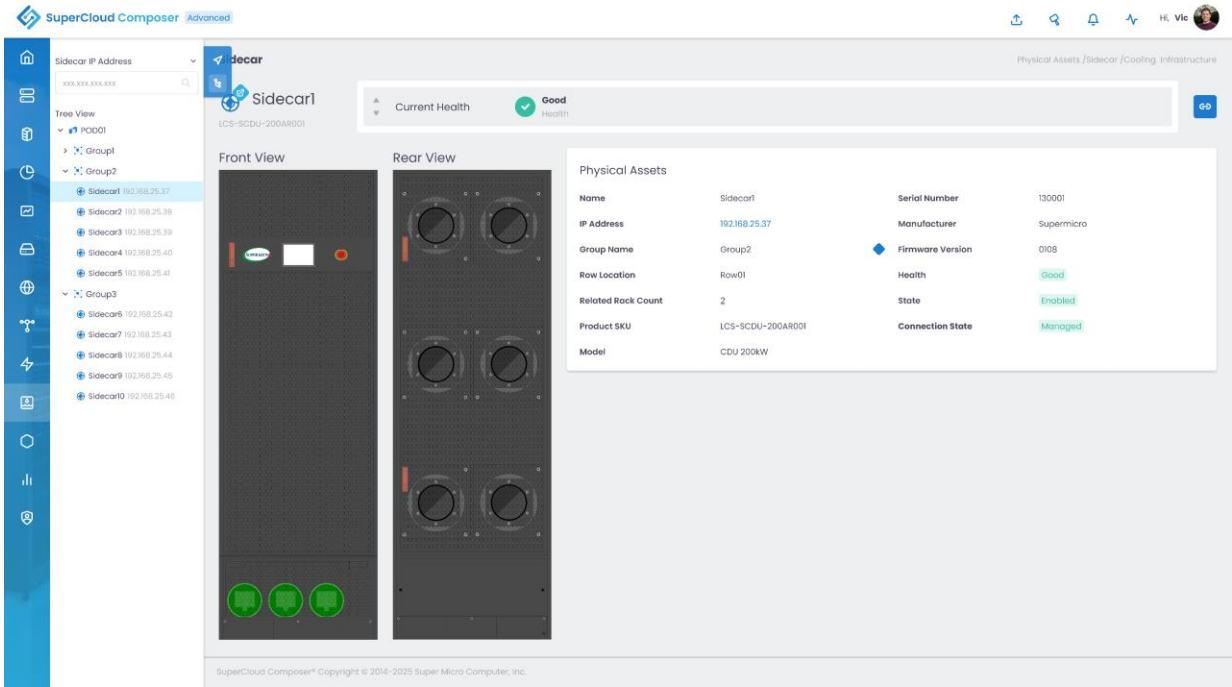


Figure 45. Cooling Infrastructure – Sidecar Details & State

Rear-door rack-mounted heat exchangers (RDHx) remove server exhaust heat at the rack level and, when integrated with monitoring systems, provide visibility into key temperatures (air outlet and water circuit points) and device health/operational state.

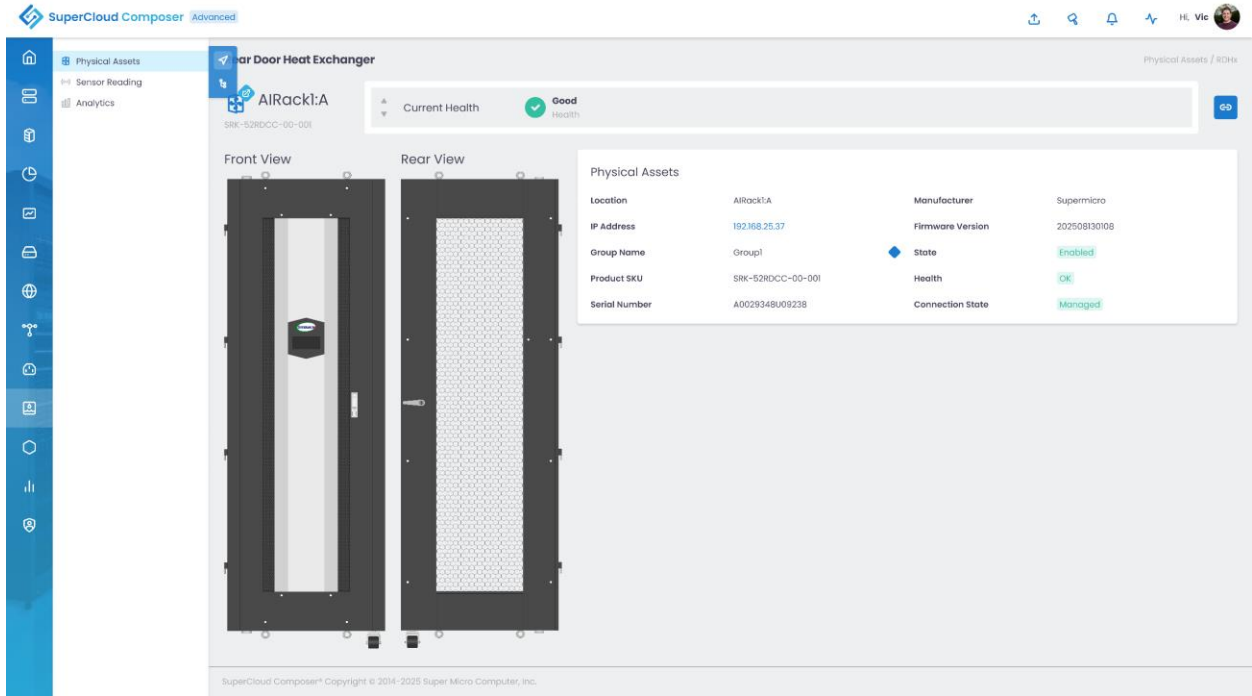


Figure 46. Cooling Infrastructure – Rear Door Heat Exchanger Details & State

- Leak detection with policy enforcement, supporting automated protection actions and coordinated safety responses on detection events.

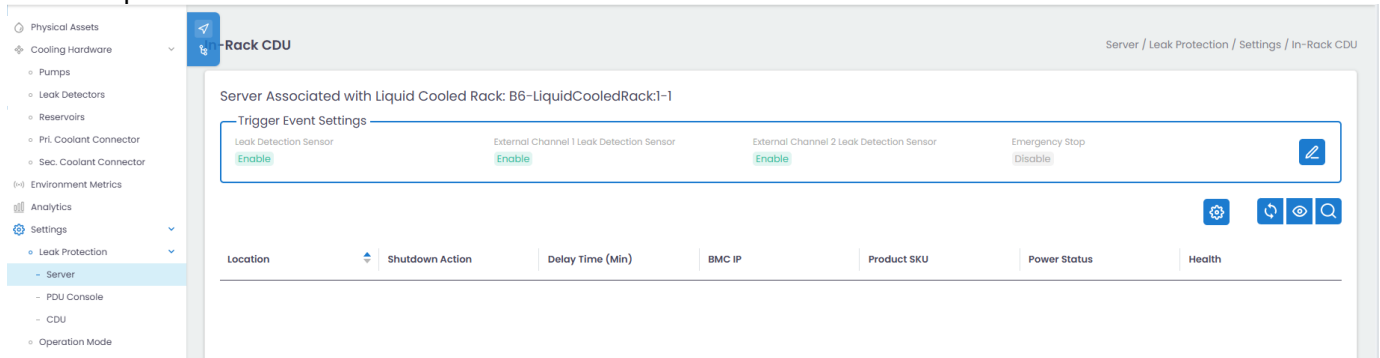


Figure 47. Cooling Infrastructure – Leak Protection Policy Actions

- Real-time cooling telemetry for coolant temperature, pressure, flow rate, humidity, power draw, and pump/valve state.

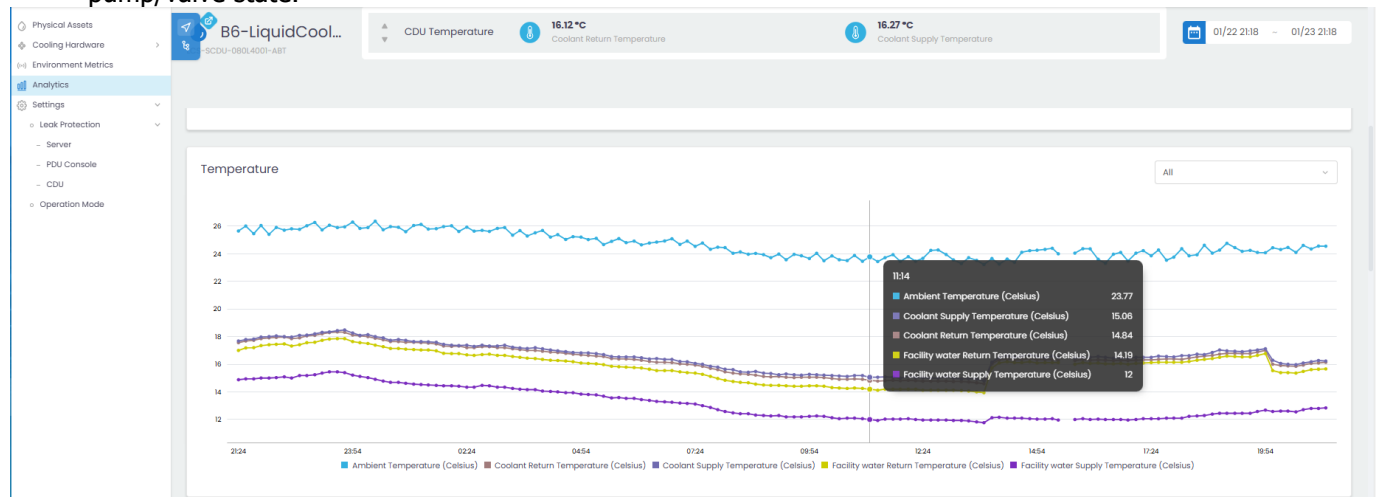


Figure 48. Cooling Infrastructure – Detailed Analytics Cooling Temperatures



Figure 49. Cooling Infrastructure – Historical Time-series Analytics, Trend Analysis, Fault Investigation & Performance Validation

- Controlled Operating Mode and Safety control action handling, including leak-triggered protection actions and emergency stop mechanisms.

Operation Mode Settings of B6-LiquidCooledRack1-1

[Reminder]
The pump's operation will be controlled automatically based on the target flow rate while the valve's operation will be controlled automatically by current dew point settings.

Operation Mode ☒ Auto ☐ Stop

Coolant Supply Temperature °C

Coolant Flow Rate LPM

Save

Figure 50. Cooling Infrastructure – CDU Cooling Control Configuration (Flow Rate & Supply Temperature)

- Operational power control integration, coordinating cooling system actions with server and PDU power operations.

Physical Assets / Cooling Hardware / Pumps / Leak Detectors / Reservoirs / P/L Coolant Connector / Sec. Coolant Connector / Environment Metrics / Analytics / Settings / Leak Protection / Server / PDU Console / CDU / Operation Mode

PDU Console / Leak Protection / Settings / In-Rack CDU

Server Associated with Liquid Cooled Rack: B6-LiquidCooledRack1-1

Trigger Event Settings

Leak Detection Sensor ☐ Disable

Show 10 Entries

☒ System Drawer

☒ B6-LiquidCooledRack1-1

Showing 1 to 1 of 1 Entries

PDU Enforcement Setting
Set Outlet Enforcement policy.

[Caution]

- Enforce PDU Outlet action to standby power for servers impacted by leak.
- If the AC OFF action is executed before the Graceful Shutdown action due to the delay time settings, the system's Graceful Shutdown process may not be fully completed, or it might not execute at all.

Location	Outlet Action	Delay Time (Min)
B6-LiquidCooledRack1-1	No Action	

Save

Figure 51. Cooling Infrastructure – In-Rack PDU Policy Action Configuration

- CDU firmware lifecycle management, supporting version visibility and controlled firmware updates across supported cooling units.

Appliance IP Address 100.100.100.100

Physical Assets / In-Rack CDU

In-Rack CDU

Show 10 Entries

Location	IP Address	Group
B6-LiquidCooledRack1-1	172.24.169.17	LiquidCooledRack1-1
LiquidCooledRack1-1	172.24.169.175	LiquidCooledRack1-1

Showing 1 to 2 of 2 Entries

Update CDU
After CDU firmware flash server will perform reboot operation.

Current

IP	Location	SKU	Current version
172.24.169.17	B6-LiquidCooledRack1-1	LCS-SCDU-080L4001-ABT	1500

Select image on server

Update

Figure 52. Cooling Infrastructure – In-Rack CDU Firmware Update

PowerAsset Orchestration (New)

The PowerAsset Orchestration module provides centralized visibility and control over data-center power infrastructure across both grid-connected and behind-the-meter environments. It interfaces with utility power ingress and electrical systems to collect real-time energy consumption telemetry, while also enabling active control of integrated assets, including PDUs, Battery Energy Storage Systems (BESS), and generators.

This enables autonomous orchestration of power and energy resources to meet data center operational requirements.

- Continuous PUE insights enable intelligent orchestration of cooling systems, load shedding, and AI scheduling. Optimizes efficiency, lowers operating costs, and ensures sustainable datacenter performance.
- Direct integration of real-time energy market prices into operations. The system continuously evaluates cost-per-kWh, identifying spikes or low-cost windows.
- AI-driven business rules dynamically adjust compute loads—optimizing training jobs, shifting workloads, or ramping resources—to capture savings without disrupting operations. Enables low-latency, autonomous decision-making at the edge, rack, or data center level.
- Dynamically scales Nvidia, AMD, or Intel GPU workloads up or down based on real-time power availability and cost.
- Intelligent scheduling balances performance with energy cost while honoring contractual guarantees.



Figure 53. Power Asset Orchestration - GPU Power Telemetry with Policy-Based Power Capping

Artifacts

The Artifacts module provides a centralized, governed repository for managing firmware and system artifacts across the entire datacenter stack. It enables controlled lifecycle management of BIOS, BMC, GPU, switch, liquid-cooling, and cluster firmware, ensuring consistency, traceability, and operational safety at scale.

By decoupling artifact management from deployment workflows, Artifacts allows operators to standardize, validate, and reuse trusted images across environments, reducing risk while accelerating large-scale provisioning and lifecycle operations.

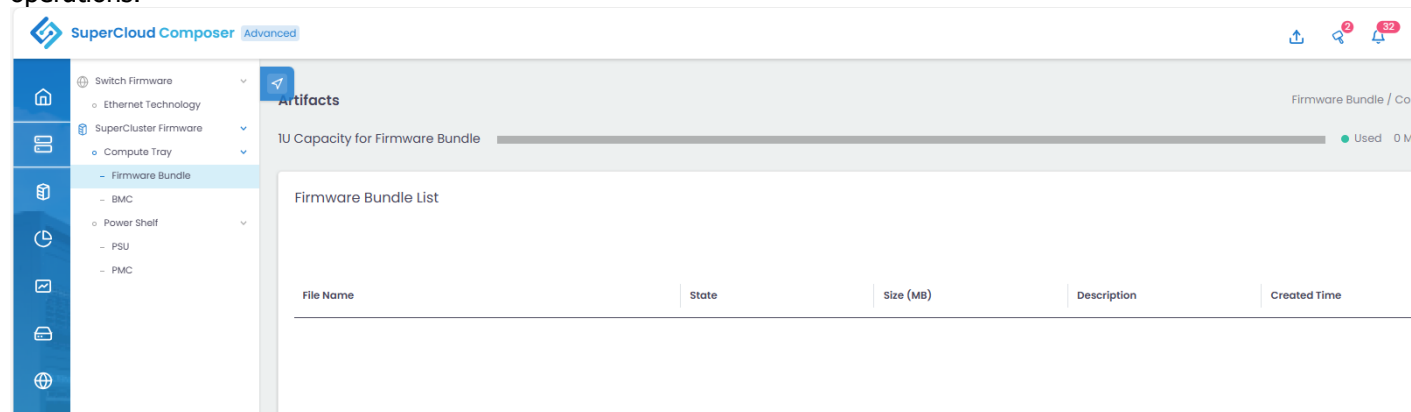


Figure 54. Artifacts Module – Firmware Repository List

Log

The Logs module provides a unified, auditable logging framework across IT, infrastructure, and facility domains, delivering end-to-end visibility into datacenter operations. It aggregates and normalizes logs from BMCs, GPUs, PDUs, in-rack and in-row CDUs, cooling towers, and platform workflows, alongside user activities and system tasks, enabling consistent operational insight from hardware to orchestration layers.

Logs are time-indexed, severity-classified, searchable, and exportable, supporting root-cause analysis, compliance reporting, and operational governance. Fine-grained filtering, role-based visibility, and lifecycle controls (search, export, retention, deletion) ensure logs can be consumed by operators, administrators, and auditors while maintaining platform security and accountability.

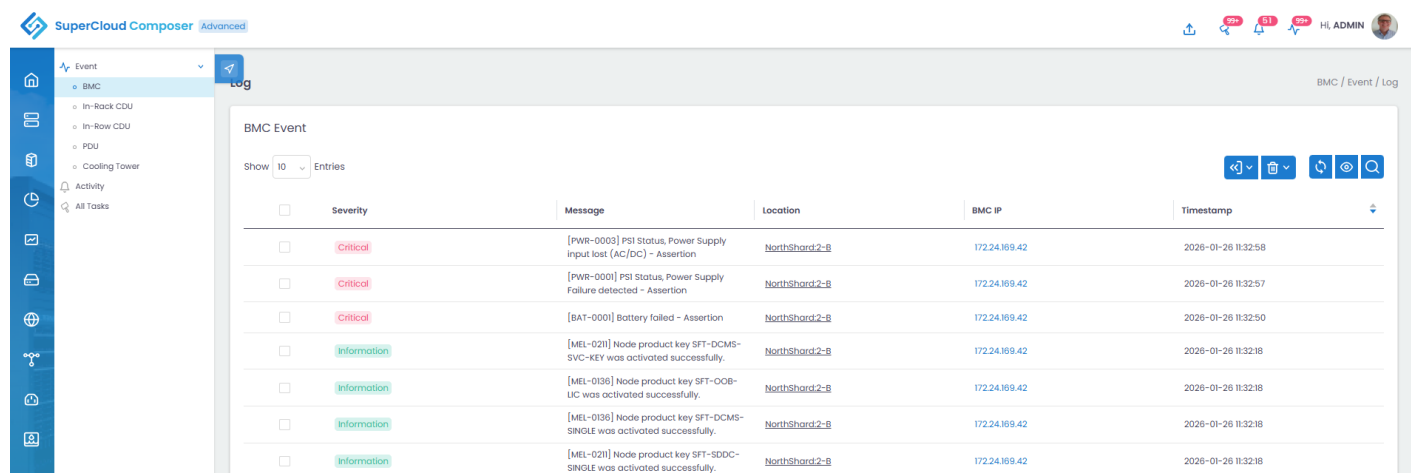


Figure 55. Log Module – All Infrastructure Domain Event Logs & Task Activity Logs

ADMIN

The Admin module provides centralized governance, security, and lifecycle control for SuperCloud Composer, delivering identity-centric access control and supporting an integrated IAM layer, including Keycloak. It enforces fine-grained RBAC access and integrates with enterprise identity providers such as LDAP and Active Directory, aligned with separation-of-duties and least-privilege principles.

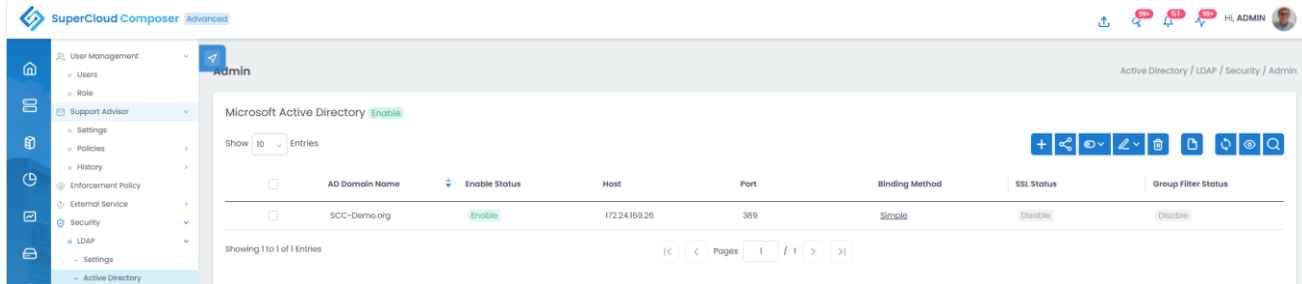


Figure 56. Admin Module – Microsoft Active Directory Integration

Administrative capabilities include tenant and project governance, user and role management, and privilege boundaries. These mechanisms allow operators to securely segment environments while maintaining centralized oversight, supporting shared-infrastructure and sovereign-domain deployment models without operational fragmentation.

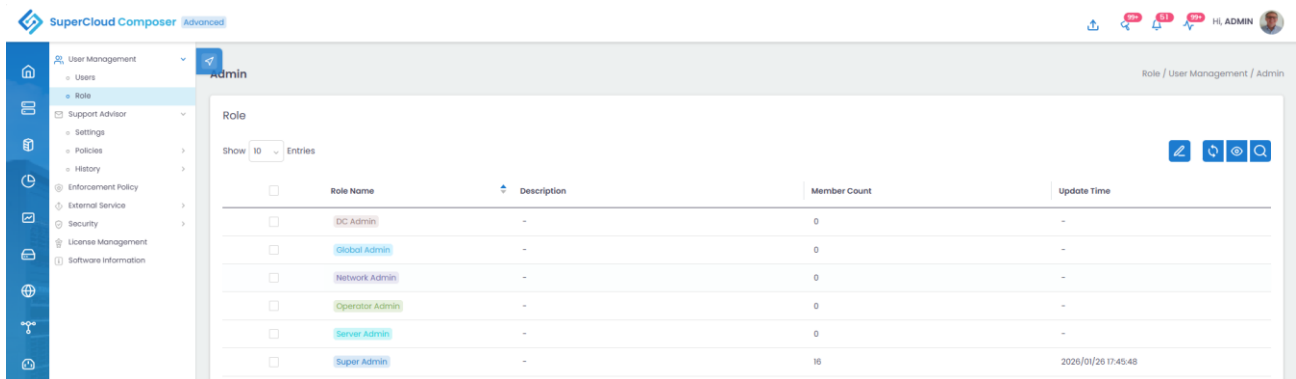


Figure 57. Admin Module – User Management & Roles

Support Advisor provides policy-driven monitoring and automated alerting across key infrastructure domains, including BMCs, PDUs, in-rack and in-row CDUs, and cooling systems. Administrators can define condition-based policies, configure notification delivery through external services such as webhooks and SMTP, assign recipients, and validate end-to-end alert workflows using test service calls.

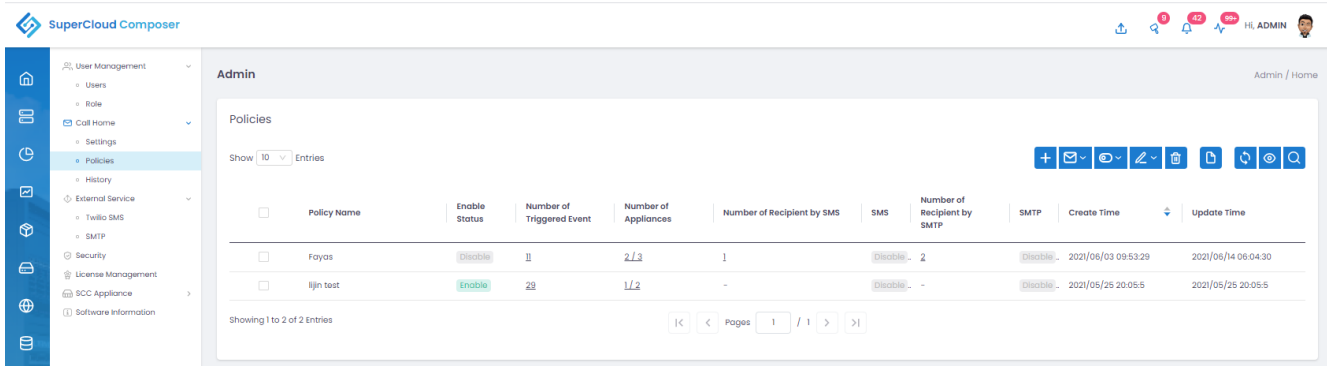


Figure 58. Admin Module – User Notification Policies

SuperCloud Composer

 H4 ADMIN

Admin / Home

History

Show 10 Entries

Alert ID	Appliance Location	Trigger Name	Trigger Severity	Event Message	Reference Policies	Number of Phone	Number of Email	Create Time
> 273		-	-	-	lin_test	0 / 0	0 / 0	2021/06/03 09:55:4
> 272		-	-	-	fsvosr	0 / 0	3 / 3	2021/06/03 09:55:4
> 271	San Jose [5-]	Power On/Off	-	-		0 / 0	0 / 0	2021/05/17 00:44:25
> 270	San Jose [5-]	Power On/Off	-	-		0 / 0	0 / 0	2021/05/17 00:18:40
> 269		Memory - Unconnecta...	Error	-	Beccolalolalola jamesdimntest	0 / 0	6 / 6	2021/05/14 10:53:14
> 268	[325-2	Power On/Off	-	-		0 / 0	0 / 0	2021/05/14 02:32:23
> 267	2228-[	GPU Card Overheat	Warning	-	Beccolalolalola	0 / 0	3 / 3	2021/05/14 02:20:41
> 266	[325-2	Power On/Off	-	-		0 / 0	0 / 0	2021/05/14 02:14:3
> 265	[325-2	Power On/Off	-	-		0 / 0	0 / 0	2021/05/14 02:13:12
> 264	[325-2	Power On/Off	-	-		0 / 0	0 / 0	2021/05/14 02:12:35

Showing 1 to 10 of 273 Entries

ServiceNow – Incident Management Integration

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Third-party device support

SuperCloud Composer is designed to integrate with heterogeneous data center environments, enabling consistent management and visibility across Supermicro and supported third-party infrastructure.

- SuperCloud Composer is built around an ODIM (Open Distributed Infrastructure Management) framework, enabling standardized, vendor-agnostic management of heterogeneous infrastructure through Redfish-aligned APIs.

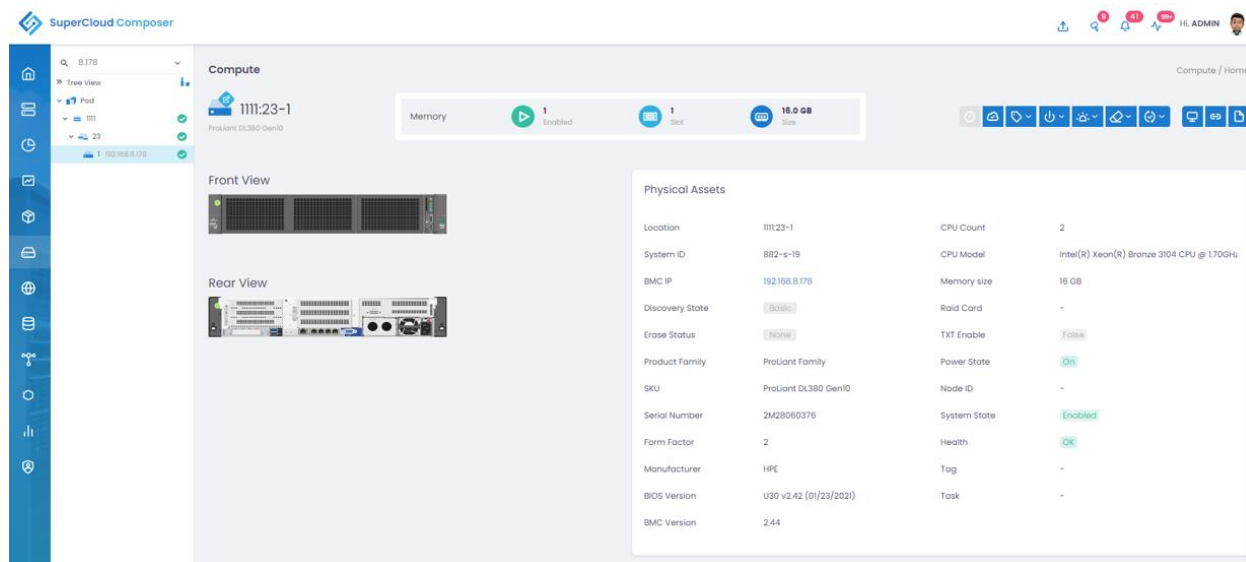


Figure 60. 3rd Party Device Management

- By leveraging open standards and industry-aligned interfaces, SCC provides a unified control plane that reduces operational silos while preserving vendor-specific capabilities.

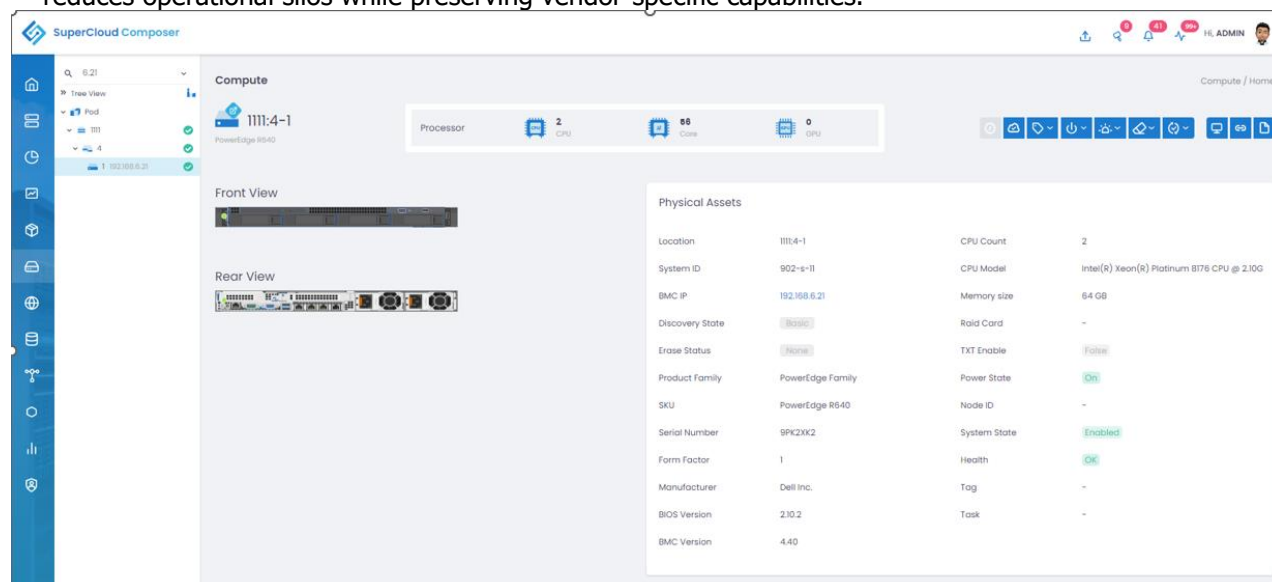


Figure 61. 3rd Party Device Management

Planned Platform Enhancements

SuperCloud Composer continues to evolve to address the operational, security, and scalability requirements of modern data center environments. The following planned enhancements reflect ongoing investments to further automate lifecycle management, improve operational visibility, and strengthen governance across distributed infrastructure.

Upcoming Firmware & Lifecycle Management Capabilities in SuperCloud Composer v3.13.

- Automated ingestion of firmware catalogs and baseline creation to maintain consistent firmware standards at scale.
- Flexible baseline definition and application across server groups, racks, Pods, product families, and data centers.
- Centralized compliance assessment with real-time status visibility and severity classification.
- Streamlined remediation workflows, including bulk and policy-driven firmware updates.
- Support for granular baselines, enabling separation of BIOS, firmware, and driver update policies, including no-reboot update paths where applicable.
- Scheduled compliance scans and update execution to reduce manual intervention and firmware drift.
- Role-based access control applied to baseline visibility and management for enterprise governance.
- Pre-update impact assessment and post-execution reporting to support operational assurance and audit requirements.
- Secure and scalable firmware update execution designed for large-scale, distributed environments.

Conclusion

SuperCloud Composer is a composable cloud management platform that provides a unified dashboard to administer software-defined data centers. Supermicro's cloud infrastructure management software brings speed, agility, and simplicity to IT administration by integrating data center tasks into a single intelligent management solution. Our robust composer engine can orchestrate cloud workloads through a streamlined industry-standard Redfish API. In addition, SuperCloud Composer monitors and manages the broad portfolio of multi-generation Supermicro servers and third-party systems through its data center lifecycle management feature set from a single unified console.

SUPERMICRO

Supermicro (Nasdaq: SMCI), the leading innovator in high-performance, high-efficiency server and storage technology is a premier provider of advanced server Building Block Solutions® for Enterprise Data Center, Cloud Computing, Artificial Intelligence, and Edge Computing Systems worldwide. Supermicro is committed to protecting the environment through its “We Keep IT Green®” initiative and provides customers with the most energy-efficient, environmentally friendly solutions available on the market.